

My Project

Generated by Doxygen 1.14.0

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Chapter 1

Data Structure Index

1.1 Data Structures

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Chapter 2

File Index

2.1 File List

Here is a list of all documented files with brief descriptions:

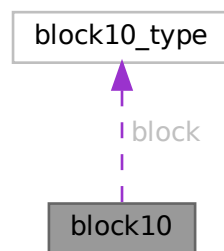
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Chapter 3

Data Structure Documentation

3.1 block10 Union Reference

Collaboration diagram for block10:



Data Fields

- `uint8_t b [sizeof(block10\_type)]`
- `block10\_type block`

The documentation for this union was generated from the following file:

- `gemsecs.h`

3.2 block10_type Struct Reference

Data Fields

- uint32_t **systemb**
- uint8_t **bidl**
- uint8_t **bidh**: 7
- uint8_t **ebit**: 1
- uint8_t **function**
- uint8_t **stream**: 7
- uint8_t **wbit**: 1
- uint8_t **didl**
- uint8_t **didh**: 7
- uint8_t **rbit**: 1

The documentation for this struct was generated from the following file:

- gemsecs.h

3.3 bytes2 Union Reference

Data Fields

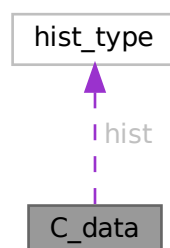
- uint16_t **ld**
- uint8_t **bd** [2]

The documentation for this union was generated from the following file:

- daq.c

3.4 C_data Struct Reference

Collaboration diagram for C_data:



Data Fields

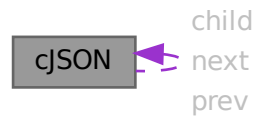
- float **calc** [ADC_BUFFER_SIZE]
- float **c_load**
- float **c_bat**
- float **c_pv**
- float **c_mppt**
- float **v_cc**
- float **v_pv**
- float **v_bat**
- float **v_cbus**
- float **v_bbat**
- float **v_temp**
- float **v_inverter**
- float **v_sensor**
- float **v_cmode**
- float **bv_ror**
- float **bc_ror**
- float **p_load**
- float **p_inverter**
- float **p_pv**
- float **p_bat**
- float **p_mppt**
- float **start_power**
- float **t_comp**
- float **esr**
- float **bank_ah**
- float **dynamic_ah**
- float **pv_ah**
- float **loadah**
- float **dynamic_ah_adj**
- float **dynamic_ah_daily**
- float **dynamic_ah_adj_daily**
- float **bkwi**
- float **bkwo**
- float **pvkw**
- float **invkw**
- float **loadkw**
- uint16_t **runtime**
- uint16_t **soc**
- bool **update**
- [hist_type](#) **hist** [1]
- float **load_i1**
- float **load_i2**
- float **bv_noload**
- float **bv_one_load**
- float **bv_full_load**
- bool **day**
- bool **dupdate**
- time_t **day_start**
- time_t **day_end**
- time_t **day_update**
- time_t **updates**

The documentation for this struct was generated from the following file:

- mbmc.h

3.5 cJSON Struct Reference

Collaboration diagram for cJSON:



Data Fields

- struct cJSON * **next**
- struct cJSON * **prev**
- struct cJSON * **child**
- int **type**
- char * **valuestring**
- int **valueint**
- double **valuedouble**
- char * **string**

The documentation for this struct was generated from the following file:

- cJSON/cJSON.h

3.6 cJSON_Hooks Struct Reference

Public Member Functions

- void *CJSON_CDECL * **malloc_fn** (size_t sz)
- void (CJSON_CDECL ***free_fn**)(void *ptr)

The documentation for this struct was generated from the following file:

- cJSON/cJSON.h

3.7 CRC_OBJ Struct Reference

Data Fields

- uint8_t **dataWidth**
- uint8_t **polyWidth**
- uint8_t **seedDirection**

3.7.1 Detailed Description

CRC Generated Driver File

@Company Microchip Technology Inc.

@File Name crc.c

@Summary This is the generated driver implementation file for the CRC driver using PIC10 / PIC12 / PIC16 / PIC18 MCUs

@Description This source file provides APIs for CRC. Generation Information : Product Revision : PIC10 / PIC12 / PIC16 / PIC18 MCUs - 1.81.8 Device : PIC18F57K42 Driver Version : 2.11 The generated drivers are tested against the following: Compiler : XC8 2.36 and above MPLAB : MPLAB X 6.00 Section: Included Files Section: Structures

The documentation for this struct was generated from the following file:

- mcc_generated_files/crc.c

3.8 D_data Struct Reference

Data Fields

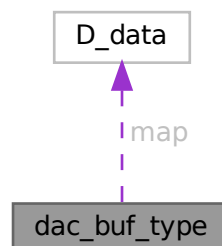
- uint8_t **dac0**: 8
- uint8_t **dac1**: 4
- uint8_t **cont**: 4
- char **lcd** [4][4][40]
- uint8_t **vterm**: 1
- D_CODES **last_info**

The documentation for this struct was generated from the following files:

- daq.c
- mydisplay.h

3.9 dac_buf_type Union Reference

Collaboration diagram for dac_buf_type:



Data Fields

- uint16_t **ld**
- uint8_t **bd** [2]
- struct [D_data](#) **map**

The documentation for this union was generated from the following file:

- daq.c

3.10 error Struct Reference

Data Fields

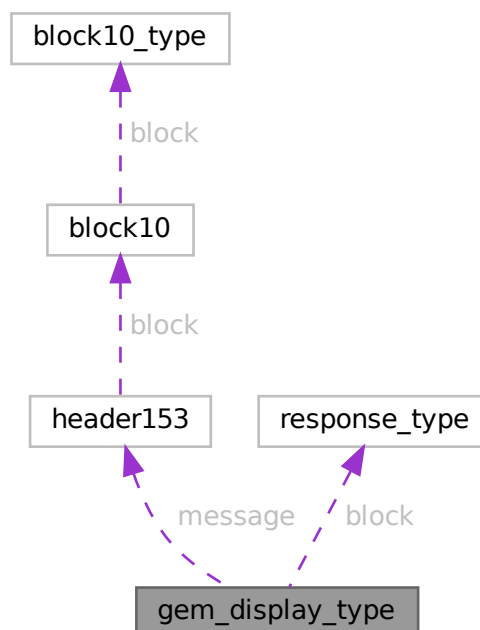
- const unsigned char * **json**
- size_t **position**

The documentation for this struct was generated from the following file:

- cJSON/cJSON.c

3.11 gem_display_type Struct Reference

Collaboration diagram for `gem_display_type`:



Data Fields

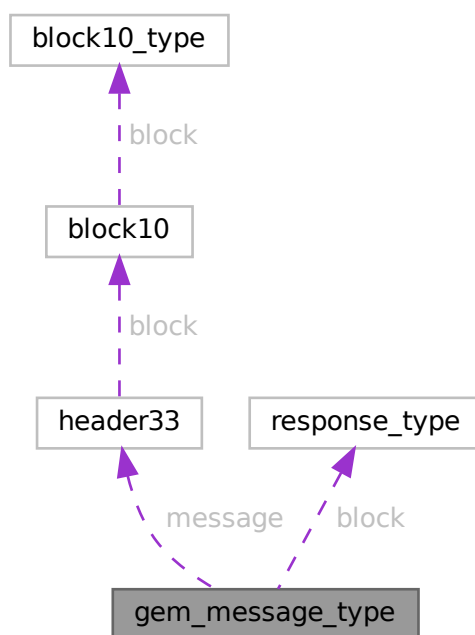
- [header153](#) **message**
- [response_type](#) **block**
- uint16_t **delay**
- uint8_t **stack**

The documentation for this struct was generated from the following file:

- gemsecs.h

3.12 gem_message_type Struct Reference

Collaboration diagram for gem_message_type:



Data Fields

- [header33](#) **message**
- [response_type](#) **block**
- uint16_t **delay**
- uint8_t **stack**

The documentation for this struct was generated from the following file:

- gemsecs.h

3.13 H_data Struct Reference

Data Fields

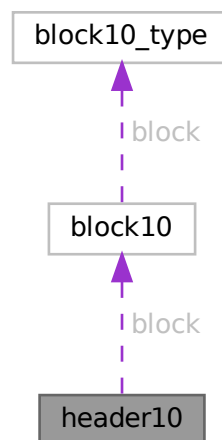
- H_CODES **hid_display**
- H_STATES **h_state**
- bool **wait_select**
- bool **select_p**
- bool **wait_enter**
- bool **enter_p**
- bool **wait_calib**
- bool **calib_p**
- uint8_t **sequence**

The documentation for this struct was generated from the following file:

- hid.h

3.14 header10 Struct Reference

Collaboration diagram for header10:



Data Fields

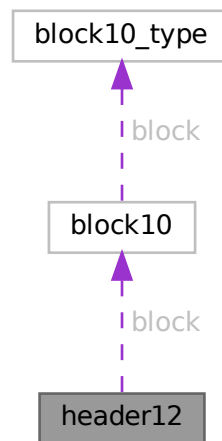
- uint16_t **checksum**
- **block10** **block**
- uint8_t **length**

The documentation for this struct was generated from the following file:

- gemsecs.h

3.15 header12 Struct Reference

Collaboration diagram for header12:



Data Fields

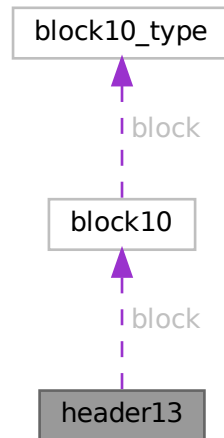
- `uint16_t` **checksum**
- `uint8_t` **data** [2]
- `block10` **block**
- `uint8_t` **length**

The documentation for this struct was generated from the following file:

- `gemsecs.h`

3.16 header13 Struct Reference

Collaboration diagram for header13:



Data Fields

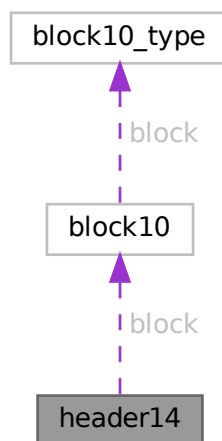
- `uint16_t` **checksum**
- `uint8_t` **data** [3]
- `block10` **block**
- `uint8_t` **length**

The documentation for this struct was generated from the following file:

- `gemsecs.h`

3.17 header14 Struct Reference

Collaboration diagram for header14:



Data Fields

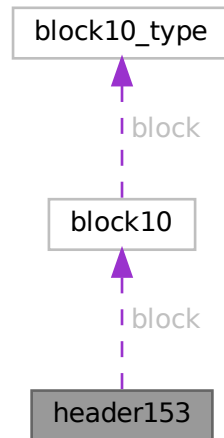
- `uint16_t` **checksum**
- `uint8_t` **data** [4]
- `block10` **block**
- `uint8_t` **length**

The documentation for this struct was generated from the following file:

- `gemsecs.h`

3.18 header153 Struct Reference

Collaboration diagram for header153:



Data Fields

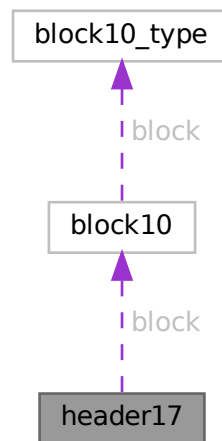
- `uint16_t` **checksum**
- `uint8_t` **data** [143]
- `block10` **block**
- `uint8_t` **length**

The documentation for this struct was generated from the following file:

- `gemsecs.h`

3.19 header17 Struct Reference

Collaboration diagram for header17:



Data Fields

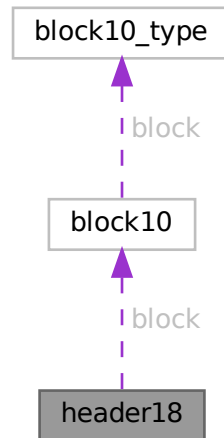
- `uint16_t` **checksum**
- `uint8_t` **data** [7]
- `block10` **block**
- `uint8_t` **length**

The documentation for this struct was generated from the following file:

- `gemsecs.h`

3.20 header18 Struct Reference

Collaboration diagram for header18:



Data Fields

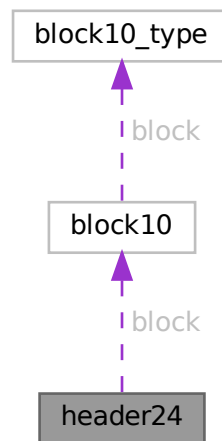
- `uint16_t` **checksum**
- `uint8_t` **data** [8]
- `block10` **block**
- `uint8_t` **length**

The documentation for this struct was generated from the following file:

- `gemsecs.h`

3.21 header24 Struct Reference

Collaboration diagram for header24:



Data Fields

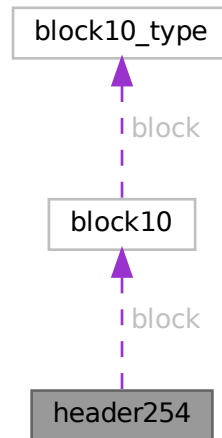
- `uint16_t checksum`
- `uint8_t data [12]`
- `uint8_t datam [2]`
- `block10 block`
- `uint8_t length`

The documentation for this struct was generated from the following file:

- `gemsecs.h`

3.22 header254 Struct Reference

Collaboration diagram for header254:



Data Fields

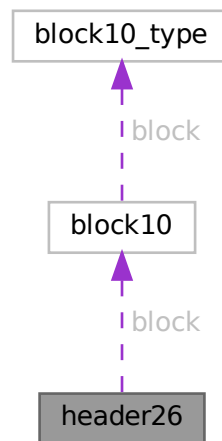
- `uint16_t` **checksum**
- `uint8_t` **data** [244]
- `block10` **block**
- `uint8_t` **length**

The documentation for this struct was generated from the following file:

- `gemsecs.h`

3.23 header26 Struct Reference

Collaboration diagram for header26:



Data Fields

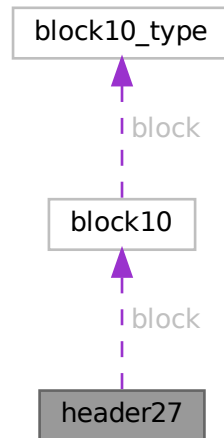
- `uint16_t checksum`
- `uint8_t data [14]`
- `uint8_t datam [2]`
- `block10 block`
- `uint8_t length`

The documentation for this struct was generated from the following file:

- `gemsecs.h`

3.24 header27 Struct Reference

Collaboration diagram for header27:



Data Fields

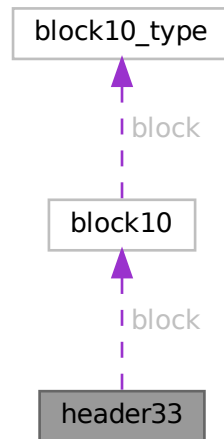
- uint16_t **checksum**
- uint8_t **data** [17]
- [block10](#) **block**
- uint8_t **length**

The documentation for this struct was generated from the following file:

- gemsecs.h

3.25 header33 Struct Reference

Collaboration diagram for header33:



Data Fields

- `uint16_t` **checksum**
- `uint8_t` **data** [23]
- `block10` **block**
- `uint8_t` **length**

The documentation for this struct was generated from the following file:

- `gemsecs.h`

3.26 hist_type Struct Reference

Data Fields

- `int32_t` **pv_eff**
- `int32_t` **tot_eff**
- `uint32_t` **ttg_t**
- `uint32_t` **updates**
- `time_t` **pclock**
- `float` **peukert**
- `float` **cef**
- `float` **peukert_adj**
- `float` **cef_calc**
- `float` **cef_save**

- `int16_t h` [HPARAM_SIZE]
- `uint16_t rate`
- `uint16_t udod`
- `uint16_t bsoc`
- `uint16_t bound_rate`
- `uint16_t bound_factor`
- `uint16_t samplei`
- `uint16_t sampleo`
- `uint16_t ah`
- `uint16_t drate`
- `uint16_t esr`
- `uint16_t rest_rate`
- `uint16_t rest_factor`
- `uint16_t esrp`
- `uint8_t version`

The documentation for this struct was generated from the following file:

- `vconfig.h`

3.27 internal_hooks Struct Reference

Public Member Functions

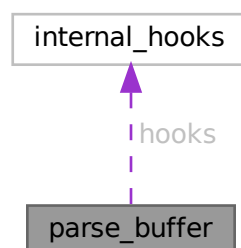
- `void *CJSON_CDECL * allocate` (`size_t` size)
- `void` (`CJSON_CDECL *deallocate`)(`void *pointer`)
- `void *CJSON_CDECL * reallocate` (`void *pointer`, `size_t` size)

The documentation for this struct was generated from the following file:

- `cjson/cJSON.c`

3.28 parse_buffer Struct Reference

Collaboration diagram for `parse_buffer`:



Data Fields

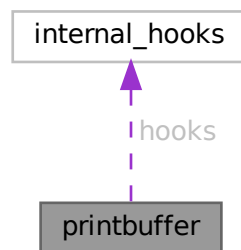
- const unsigned char * **content**
- size_t **length**
- size_t **offset**
- size_t **depth**
- [internal_hooks](#) **hooks**

The documentation for this struct was generated from the following file:

- cJSON/cJSON.c

3.29 printbuffer Struct Reference

Collaboration diagram for printbuffer:



Data Fields

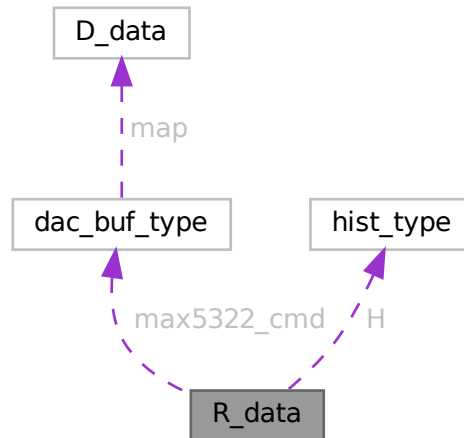
- unsigned char * **buffer**
- size_t **length**
- size_t **offset**
- size_t **depth**
- cJSON_bool **noalloc**
- cJSON_bool **format**
- [internal_hooks](#) **hooks**

The documentation for this struct was generated from the following file:

- cJSON/cJSON.c

3.30 R_data Struct Reference

Collaboration diagram for R_data:



Data Fields

- `adc_result_t` **raw_adc** [ADC_BUFFER_SIZE]
- `adc_result_t` **raw_dac** [DAC_BUFFER_SIZE]
- `union` `dac_buf_type` **max5322_cmd**
- `int16_t` **n_offset** [NUM_C_SENSORS]
- `float` **n_scalar** [NUM_C_SENSORS]
- `uint8_t` **scan_index**
- `uint16_t` **scan_select**
- `bool` **done**
- `hist_type` **H**
- `uint8_t` **hist_save**: 1
- `uint8_t` **hist_update**: 1
- `uint8_t` **c_zero_cal**: 1
- `uint8_t` **c_scale_cal**: 1
- `uint16_t` **checkmark**
- `uint8_t` **crc**

The documentation for this struct was generated from the following file:

- `daq.c`

3.31 rbutton_type Struct Reference

Data Fields

- SW_STATES **sw**
- uint32_t **count**

The documentation for this struct was generated from the following file:

- dio.h

3.32 response_type Struct Reference

Data Fields

- uint8_t * **header**
- uint8_t **length**
- uint8_t * **reply**
- uint8_t **reply_length**
- uint8_t **respond**: 1

The documentation for this struct was generated from the following file:

- gemsecs.h

3.33 ringBufS_t Struct Reference

Data Fields

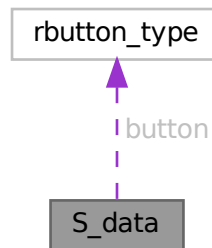
- uint8_t **buf** [RBUF_SIZE]
- uint8_t **head**
- uint8_t **tail**
- uint8_t **count**

The documentation for this struct was generated from the following file:

- ringbufs.h

3.34 S_data Struct Reference

Collaboration diagram for S_data:



Data Fields

- volatile `rbutton_type` `button` [MAX_BUTTONS]
- volatile `uint8_t` `sw_bitmap`

The documentation for this struct was generated from the following file:

- `dio.c`

3.35 spi1_configuration_t Struct Reference

Data Fields

- `uint8_t` `con0`
- `uint8_t` `con1`
- `uint8_t` `con2`
- `uint8_t` `baud`
- `uint8_t` `operation`

3.35.1 Detailed Description

SPI1 Generated Driver File

@Company Microchip Technology Inc.

@File Name spi1.c

@Summary This is the generated driver implementation file for the SPI1 driver using PIC10 / PIC12 / PIC16 / PIC18 MCUs

@Description This header file provides implementations for driver APIs for SPI1. Generation Information : Product Revision : PIC10 / PIC12 / PIC16 / PIC18 MCUs - 1.81.8 Device : PIC18F57K42 Driver Version : 1.0.0 The generated drivers are tested against the following: Compiler : XC8 2.36 and above or later MPLAB : MPLAB X 6.00

The documentation for this struct was generated from the following file:

- `mcc_generated_files/spi1.c`

3.36 spi_link_type Struct Reference

Data Fields

- uint8_t **SPI_LCD**: 1
- uint8_t **SPI_AUX**: 1
- uint8_t **LCD_TIMER**: 1
- volatile uint8_t **LCD_DATA**: 1
- uint16_t **delay**
- uint8_t **config**
- uint8_t * **txbuf**
- volatile int32_t **int_count**

The documentation for this struct was generated from the following file:

- vconfig.h

3.37 terminal_type Struct Reference

Data Fields

- uint8_t **mesgid**
- uint8_t **TID**
- uint8_t **mcode**
- uint8_t **mparm**
- uint8_t **cmdlen**
- uint8_t **log_seq**
- uint8_t **host_display_ack**: 1
- D_CODES **info**
- D_CODES **help_temp**
- uint16_t **ceid**
- uint16_t **log_num**

The documentation for this struct was generated from the following file:

- vconfig.h

3.38 uart1_status_t Union Reference

```
#include <uart1.h>
```

Data Fields

- struct {
 unsigned **perr**: 1
 unsigned **ferr**: 1
 unsigned **oerr**: 1
 unsigned **reserved**: 5
};
- uint8_t **status**

3.38.1 Detailed Description

Section: Data Type Definitions

The documentation for this union was generated from the following file:

- `mcc_generated_files/uart1.h`

3.39 `uart2_status_t` Union Reference

```
#include <uart2.h>
```

Data Fields

- struct {
 - unsigned **perr**: 1
 - unsigned **ferr**: 1
 - unsigned **oerr**: 1
 - unsigned **reserved**: 5
- };
- `uint8_t status`

3.39.1 Detailed Description

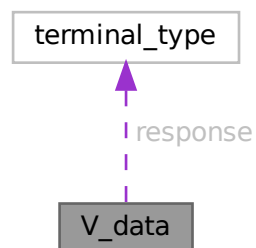
Section: Data Type Definitions

The documentation for this union was generated from the following file:

- `mcc_generated_files/uart2.h`

3.40 `V_data` Struct Reference

Collaboration diagram for `V_data`:



Data Fields

- UI_STATES **ui_state**
- char **buf** [64]
- char **info** [64]
- char **rbuf** [16]
- volatile time_t **ticks**
- volatile time_t **blight**
- volatile time_t **ac_time**
- volatile int8_t **testing**
- volatile uint8_t **sys_info**: 1
- volatile uint8_t **ac_on**: 1
- volatile uint8_t **ac_off**: 1
- volatile uint8_t **time_info**: 1
- volatile uint8_t **get_time_text**: 1
- uint8_t **error**
- uint8_t **abort**
- uint8_t **msg_error**
- uint8_t **msg_ret**
- uint8_t **alarm**
- uint8_t **float_ticks**
- uint8_t **boost_ticks**
- uint8_t **wdt_ticks**
- UI_STATES **ui_sw**
- uint16_t **r_checksum**
- uint16_t **t_checksum**
- uint16_t **checksum_error**
- uint16_t **mode_pwm**
- uint16_t **sequences**
- uint16_t **all_errors**
- uint8_t **set_sequ**: 1
- uint8_t **debug**: 1
- uint8_t **help**: 1
- uint8_t **stack**: 3
- uint8_t **help_id**: 2
- uint8_t **screen**: 1
- [terminal_type](#) response
- volatile uint8_t **ticker**
- volatile uint8_t **cc_state**
- bool **flipper**
- bool **calib**
- bool **enter**
- bool **sensor_set**
- bool **fixup**
- bool **cc_active**
- volatile bool **system_stable**
- volatile bool **blight_off**
- volatile bool **in_float**
- volatile bool **in_boost**
- volatile uint32_t **lowint_count**
- volatile uint32_t **timerint_count**
- volatile uint32_t **tx_count**
- volatile uint32_t **rx_count**
- volatile uint32_t **adc_count**
- volatile uint32_t **spi_count**

- volatile uint32_t **switch_count**

The documentation for this struct was generated from the following file:

- vconfig.h

3.41 V_help Struct Reference

Data Fields

- const char **message** [22]
- const char **display** [22]

The documentation for this struct was generated from the following file:

- vconfig.h

Chapter 4

File Documentation

4.1 bsoc.h

```
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00017 *
00018 * MICROCHIP PROVIDES THIS SOFTWARE CONDITIONALLY UPON YOUR ACCEPTANCE OF THESE
00019 * TERMS.
00020 */
00021
00022 /*
00023 * File:
00024 * Author:
00025 * Comments:
00026 * Revision history:
00027 */
00028
00029 // This is a guard condition so that contents of this file are not included
00030 // more than once.
00031 #ifndef BSOC_H
00032 #define BSOC_H
00033
00034 #include <xc.h> // include processor files - each processor file is guarded.
00035 #include <math.h>
00036 #include <stdio.h>
00037 #include <string.h>
00038 #include "mcc_generated_files/tmr3.h"
00039 #include "vconfig.h"
00040 #include "mbmc.h"
00041 #include "eadog.h"
00042 #include "timers.h"
00043 #include "hid.h"
00044 #include "mconfig.h"
00045
00046 #define FM80_FIXUPS // correct current variables
00047
00048 #define MAXRUNTIME 95.0f
00049 #define MINRUNTIME 0.05f
00050 #define Perk_ADJ_FL 1.10f // If the peukert_adj is below this, adjust the Ah out by this Ah
00051 // adjustment factor
00052 #define Perk_ADJ_FH 5.00f // If the peukert_adj is above this, adjust the Ah out by this Ah
00053 // adjustment factor
00054 #define Temp_ADJ_FL 1.20f // Adjust the battery cap up or down 1C by this percentage from Temp_ZERO
00055 #define Temp_ZERO 2671 // Temperature for base battery rate capacity in tenths of C
00056 #define CEF_HSOC 0.45f // Charge Efficiency Factor, high SOC
00057 #define CEF_MSOC 0.70f // middle
```

```

00056 #define CEF_LSOC      0.90f    // Charge Efficiency Factor, low SOC
00057 #define CEF_OVER       0.99f    // CEF overflow in calc
00058 #define CEF_MAX        0.95f    // Invalid charge factor above this, should be about the same as CEF_LOSC
00059 #define CEF_MIN        0.40f    // Invalid charge factor below this
00060 #define CEF_RESET      0.80f    // reset AH values below this point.
00061 #define CEF_DERATE     1.20f    // Ahi derate factor to start dynamic CEF corrections
00062 #define SOC_DERATE     70        // derate CEF above this
00063 #define SOC_CEF        100 // Max state of charge to update dynamic charge efficiency factors
00064 #define HI_ESR         1500     // highest possible esr
00065
00066 #define BVSOC_SLOTS    26        // Battery to SOC data table slots
00067
00068 #define FLOAT_TIME     60 // cc in float mode delay time before flag set
00069
00070 /*
00071  * PV side diversion
00072  */
00073 #define DPWM_FULL      300
00074 #define DPWM_LOW_VOLTS 40.0f
00075 #define PV_MAX_POWER   300.0f
00076 #define PWM_MAX_POWER  250.0f
00077 #define PWM_WIN_POWER  10.0f
00078
00079 #define GTI_MAX        600 // 600
00080 #define GTI_IDLE       50
00081 #define GTI_INCR       100
00082 #define GTI_NORM       450 // 450
00083
00084 typedef enum {
00085     R_CYCLE = 0,
00086     R_TOTAL,
00087     R_PV,
00088     R_BAT,
00089 } R_CODES;
00090
00091 typedef enum {
00092     M_DEACT = 0,
00093     M_ACT = 1,
00094     M_MPPT = 2,
00095     M_EQUAL = 3,
00096     M_BOOST = 4,
00097     M_FLOAT = 5,
00098     M_LIMIT = 6,
00099     M_OFFLINE = 255,
00100 } M_CODES;
00101
00102 extern volatile C_data C;
00103 extern V_data V;
00104 extern H_data H;
00105 extern const char *build_date, *build_time;
00106
00107 void calc_bsoc(void);
00108 void init_bsoc(void);
00109 void start_bsoc(void);
00110 void stop_bsoc(void);
00111 void reset_bsoc(R_CODES);
00112 uint16_t Volts_to_SOC(uint32_t);
00113 uint32_t peukert(uint16_t, float, float, int16_t);
00114 float esr_check(uint8_t);
00115 uint8_t cc_state(float);
00116 bool pv_diversion(bool);
00117 void gti_cmds(void);
00118
00119 #endif /* BSOC_H */
00120

```

4.2 cJSON.h

```

00001 /*
00002  Copyright (c) 2009-2017 Dave Gamble and cJSON contributors
00003
00004  Permission is hereby granted, free of charge, to any person obtaining a copy
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```



```

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00020     THE SOFTWARE.
00021 */
00022
00023 #ifndef cJSON__h
00024 #define cJSON__h
00025
00026 #ifdef __cplusplus
00027 extern "C"
00028 {
00029 #endif
00030
00031 #if !defined(__WINDOWS__) && (defined(WIN32) || defined(WIN64) || defined(_MSC_VER) ||
    defined(_WIN32))
00032 #define __WINDOWS__
00033 #endif
00034
00035 #ifdef __WINDOWS__
00036
00037 /* When compiling for windows, we specify a specific calling convention to avoid issues where we are
    being called from a project with a different default calling convention. For windows you have 3
    define options:
00038
00039 cJSON_HIDE_SYMBOLS - Define this in the case where you don't want to ever dllexport symbols
00040 cJSON_EXPORT_SYMBOLS - Define this on library build when you want to dllexport symbols (default)
00041 cJSON_IMPORT_SYMBOLS - Define this if you want to dllimport symbol
00042
00043 For *nix builds that support visibility attribute, you can define similar behavior by
00044 setting default visibility to hidden by adding
00045 -fvisibility=hidden (for gcc)
00046 or
00047 -xldscope=hidden (for sun cc)
00048 to CFLAGS
00049
00050 then using the cJSON_API_VISIBILITY flag to "export" the same symbols the way cJSON_EXPORT_SYMBOLS
    does
00052 */
00053
00054 #define cJSON_CDECL __cdecl
00055 #define cJSON_STDCALL __stdcall
00056
00057 /* export symbols by default, this is necessary for copy pasting the C and header file */
00058 #if !defined(cJSON_HIDE_SYMBOLS) && !defined(cJSON_IMPORT_SYMBOLS) && !defined(cJSON_EXPORT_SYMBOLS)
00059 #define cJSON_EXPORT_SYMBOLS
00060 #endif
00061 #endif
00062
00063 #if defined(cJSON_HIDE_SYMBOLS)
00064 #define cJSON_PUBLIC(type)   type cJSON_STDCALL
00065 #elif defined(cJSON_EXPORT_SYMBOLS)
00066 #define cJSON_PUBLIC(type)   __declspec(dllexport) type cJSON_STDCALL
00067 #elif defined(cJSON_IMPORT_SYMBOLS)
00068 #define cJSON_PUBLIC(type)   __declspec(dllimport) type cJSON_STDCALL
00069 #endif
00070 #else /* !__WINDOWS__ */
00071 #define cJSON_CDECL
00072 #define cJSON_STDCALL
00073
00074 #if (defined(__GNUC__) || defined(__SUNPRO_CC) || defined(__SUNPRO_C)) &&
    defined(cJSON_API_VISIBILITY)
00075 #define cJSON_PUBLIC(type)   __attribute__((visibility("default"))) type
00076 #else
00077 #define cJSON_PUBLIC(type)   type
00078 #endif
00079 #endif
00080
00081 /* project version */
00082 #define cJSON_VERSION_MAJOR 1
00083 #define cJSON_VERSION_MINOR 7
00084 #define cJSON_VERSION_PATCH 16
00085
00086 #include <stddef.h>
00087
00088 /* cJSON Types: */
00089 #define cJSON_Invalid (0)
00090 #define cJSON_False  (1 << 0)
00091 #define cJSON_True    (1 << 1)
00092 #define cJSON_NULL     (1 << 2)
00093 #define cJSON_Number   (1 << 3)
00094 #define cJSON_String   (1 << 4)
00095 #define cJSON_Array    (1 << 5)
00096 #define cJSON_Object   (1 << 6)
00097 #define cJSON_Raw      (1 << 7) /* raw json */

```

```

00098
00099 #define cJSON_IsReference 256
00100 #define cJSON_StringIsConst 512
00101
00102 /* The cJSON structure: */
00103 typedef struct cJSON
00104 {
00105     /* next/prev allow you to walk array/object chains. Alternatively, use
00106     GetArraySize/GetArrayItem/GetObjectItem */
00107     struct cJSON *next;
00108     struct cJSON *prev;
00109     /* An array or object item will have a child pointer pointing to a chain of the items in the
00110     array/object. */
00111     struct cJSON *child;
00112
00113     /* The type of the item, as above. */
00114     int type;
00115
00116     /* The item's string, if type==cJSON_String and type == cJSON_Raw */
00117     char *valuestring;
00118     /* writing to valueint is DEPRECATED, use cJSON_SetNumberValue instead */
00119     int valueint;
00120     /* The item's number, if type==cJSON_Number */
00121     double valuedouble;
00122
00123     /* The item's name string, if this item is the child of, or is in the list of subitems of an
00124     object. */
00125     char *string;
00126 } cJSON;
00127
00128 typedef struct cJSON_Hooks
00129 {
00130     /* malloc/free are CDECL on Windows regardless of the default calling convention of the
00131     compiler, so ensure the hooks allow passing those functions directly. */
00132     void *(CJSON_CDECL *malloc_fn)(size_t sz);
00133     void (CJSON_CDECL *free_fn)(void *ptr);
00134 } cJSON_Hooks;
00135
00136 typedef int cJSON_bool;
00137
00138 /* Limits how deeply nested arrays/objects can be before cJSON rejects to parse them.
00139 * This is to prevent stack overflows. */
00140 #ifndef cJSON_NESTING_LIMIT
00141 #define cJSON_NESTING_LIMIT 1000
00142 #endif
00143
00144 /* returns the version of cJSON as a string */
00145 cJSON_PUBLIC(const char*) cJSON_Version(void);
00146
00147 /* Supply malloc, realloc and free functions to cJSON */
00148 cJSON_PUBLIC(void) cJSON_InitHooks(cJSON_Hooks* hooks);
00149
00150 /* Memory Management: the caller is always responsible to free the results from all variants of
00151 cJSON_Parse (with cJSON_Delete) and cJSON_Print (with stdlib free, cJSON_Hooks.free_fn, or cJSON_free
00152 as appropriate). The exception is cJSON_PrintPreallocated, where the caller has full responsibility of
00153 the buffer. */
00154 /* Supply a block of JSON, and this returns a cJSON object you can interrogate. */
00155 cJSON_PUBLIC(cJSON *) cJSON_Parse(const char *value);
00156 cJSON_PUBLIC(cJSON *) cJSON_ParseWithLength(const char *value, size_t buffer_length);
00157 /* ParseWithOpts allows you to require (and check) that the JSON is null terminated, and to retrieve
00158 the pointer to the final byte parsed. */
00159 /* If you supply a ptr in return_parse_end and parsing fails, then return_parse_end will contain a
00160 pointer to the error so will match cJSON_GetErrorPtr(). */
00161 cJSON_PUBLIC(cJSON *) cJSON_ParseWithOpts(const char *value, const char **return_parse_end, cJSON_bool
00162 require_null_terminated);
00163 cJSON_PUBLIC(cJSON *) cJSON_ParseWithLengthOpts(const char *value, size_t buffer_length, const char
00164 **return_parse_end, cJSON_bool require_null_terminated);
00165
00166 /* Render a cJSON entity to text for transfer/storage. */
00167 cJSON_PUBLIC(char *) cJSON_Print(const cJSON *item);
00168 /* Render a cJSON entity to text for transfer/storage without any formatting. */
00169 cJSON_PUBLIC(char *) cJSON_PrintUnformatted(const cJSON *item);
00170 /* Render a cJSON entity to text using a buffered strategy. prebuffer is a guess at the final size.
00171 guessing well reduces reallocation. fmt=0 gives unformatted, =1 gives formatted */
00172 cJSON_PUBLIC(char *) cJSON_PrintBuffered(const cJSON *item, int prebuffer, cJSON_bool fmt);
00173 /* Render a cJSON entity to text using a buffer already allocated in memory with given length. Returns
00174 1 on success and 0 on failure. */
00175 /* NOTE: cJSON is not always 100% accurate in estimating how much memory it will use, so to be safe
00176 allocate 5 bytes more than you actually need */
00177 cJSON_PUBLIC(cJSON_bool) cJSON_PrintPreallocated(cJSON *item, char *buffer, const int length, const
00178 cJSON_bool format);
00179 /* Delete a cJSON entity and all subentities. */
00180 cJSON_PUBLIC(void) cJSON_Delete(cJSON *item);
00181
00182 /* Returns the number of items in an array (or object). */
00183 cJSON_PUBLIC(int) cJSON_GetArraySize(const cJSON *array);
00184 /* Retrieve item number "index" from array "array". Returns NULL if unsuccessful. */

```

```

00170 cJSON_PUBLIC(cJSON *) cJSON_GetArrayItem(const cJSON *array, int index);
00171 /* Get item "string" from object. Case insensitive. */
00172 cJSON_PUBLIC(cJSON *) cJSON_GetObjectItem(const cJSON * const object, const char * const string);
00173 cJSON_PUBLIC(cJSON *) cJSON_GetObjectItemCaseSensitive(const cJSON * const object, const char * const
string);
00174 cJSON_PUBLIC(cJSON_bool) cJSON_HasObjectItem(const cJSON *object, const char *string);
00175 /* For analysing failed parses. This returns a pointer to the parse error. You'll probably need to
look a few chars back to make sense of it. Defined when cJSON_Parse() returns 0. 0 when cJSON_Parse()
succeeds. */
00176 cJSON_PUBLIC(const char *) cJSON_GetErrorPtr(void);
00177
00178 /* Check item type and return its value */
00179 cJSON_PUBLIC(char *) cJSON_GetStringValue(const cJSON * const item);
00180 cJSON_PUBLIC(double) cJSON_GetNumberValue(const cJSON * const item);
00181
00182 /* These functions check the type of an item */
00183 cJSON_PUBLIC(cJSON_bool) cJSON_IsInvalid(const cJSON * const item);
00184 cJSON_PUBLIC(cJSON_bool) cJSON_IsFalse(const cJSON * const item);
00185 cJSON_PUBLIC(cJSON_bool) cJSON_IsTrue(const cJSON * const item);
00186 cJSON_PUBLIC(cJSON_bool) cJSON_IsBool(const cJSON * const item);
00187 cJSON_PUBLIC(cJSON_bool) cJSON_IsNull(const cJSON * const item);
00188 cJSON_PUBLIC(cJSON_bool) cJSON_IsNumber(const cJSON * const item);
00189 cJSON_PUBLIC(cJSON_bool) cJSON_IsString(const cJSON * const item);
00190 cJSON_PUBLIC(cJSON_bool) cJSON_IsArray(const cJSON * const item);
00191 cJSON_PUBLIC(cJSON_bool) cJSON_IsObject(const cJSON * const item);
00192 cJSON_PUBLIC(cJSON_bool) cJSON_IsRaw(const cJSON * const item);
00193
00194 /* These calls create a cJSON item of the appropriate type. */
00195 cJSON_PUBLIC(cJSON *) cJSON_CreateNull(void);
00196 cJSON_PUBLIC(cJSON *) cJSON_CreateTrue(void);
00197 cJSON_PUBLIC(cJSON *) cJSON_CreateFalse(void);
00198 cJSON_PUBLIC(cJSON *) cJSON_CreateBool(cJSON_bool boolean);
00199 cJSON_PUBLIC(cJSON *) cJSON_CreateNumber(double num);
00200 cJSON_PUBLIC(cJSON *) cJSON_CreateString(const char *string);
00201 /* raw json */
00202 cJSON_PUBLIC(cJSON *) cJSON_CreateRaw(const char *raw);
00203 cJSON_PUBLIC(cJSON *) cJSON_CreateArray(void);
00204 cJSON_PUBLIC(cJSON *) cJSON_CreateObject(void);
00205
00206 /* Create a string where valuelstring references a string so
00207 * it will not be freed by cJSON_Delete */
00208 cJSON_PUBLIC(cJSON *) cJSON_CreateStringReference(const char *string);
00209 /* Create an object/array that only references it's elements so
00210 * they will not be freed by cJSON_Delete */
00211 cJSON_PUBLIC(cJSON *) cJSON_CreateObjectReference(const cJSON *child);
00212 cJSON_PUBLIC(cJSON *) cJSON_CreateArrayReference(const cJSON *child);
00213
00214 /* These utilities create an Array of count items.
00215 * The parameter count cannot be greater than the number of elements in the number array, otherwise
array access will be out of bounds.*/
00216 cJSON_PUBLIC(cJSON *) cJSON_CreateIntArray(const int *numbers, int count);
00217 cJSON_PUBLIC(cJSON *) cJSON_CreateFloatArray(const float *numbers, int count);
00218 cJSON_PUBLIC(cJSON *) cJSON_CreateDoubleArray(const double *numbers, int count);
00219 cJSON_PUBLIC(cJSON *) cJSON_CreateStringArray(const char *const *strings, int count);
00220
00221 /* Append item to the specified array/object. */
00222 cJSON_PUBLIC(cJSON_bool) cJSON_AddItemToArray(cJSON *array, cJSON *item);
00223 cJSON_PUBLIC(cJSON_bool) cJSON_AddItemToObject(cJSON *object, const char *string, cJSON *item);
00224 /* Use this when string is definitely const (i.e. a literal, or as good as), and will definitely
survive the cJSON object.
00225 * WARNING: When this function was used, make sure to always check that (item->type &
cJSON_StringIsConst) is zero before
00226 * writing to `item->string` */
00227 cJSON_PUBLIC(cJSON_bool) cJSON_AddItemToObjectCS(cJSON *object, const char *string, cJSON *item);
00228 /* Append reference to item to the specified array/object. Use this when you want to add an existing
cJSON to a new cJSON, but don't want to corrupt your existing cJSON. */
00229 cJSON_PUBLIC(cJSON_bool) cJSON_AddItemReferenceToArray(cJSON *array, cJSON *item);
00230 cJSON_PUBLIC(cJSON_bool) cJSON_AddItemReferenceToObject(cJSON *object, const char *string, cJSON
*item);
00231
00232 /* Remove/Detach items from Arrays/Objects. */
00233 cJSON_PUBLIC(cJSON *) cJSON_DetachItemViaPointer(cJSON *parent, cJSON * const item);
00234 cJSON_PUBLIC(cJSON *) cJSON_DetachItemFromArray(cJSON *array, int which);
00235 cJSON_PUBLIC(void) cJSON_DeleteItemFromArray(cJSON *array, int which);
00236 cJSON_PUBLIC(cJSON *) cJSON_DetachItemFromObject(cJSON *object, const char *string);
00237 cJSON_PUBLIC(cJSON *) cJSON_DetachItemFromObjectCaseSensitive(cJSON *object, const char *string);
00238 cJSON_PUBLIC(void) cJSON_DeleteItemFromObject(cJSON *object, const char *string);
00239 cJSON_PUBLIC(void) cJSON_DeleteItemFromObjectCaseSensitive(cJSON *object, const char *string);
00240
00241 /* Update array items. */
00242 cJSON_PUBLIC(cJSON_bool) cJSON_InsertItemInArray(cJSON *array, int which, cJSON *newitem); /* Shifts
pre-existing items to the right. */
00243 cJSON_PUBLIC(cJSON_bool) cJSON_ReplaceItemViaPointer(cJSON * const parent, cJSON * const item, cJSON *
replacement);
00244 cJSON_PUBLIC(cJSON_bool) cJSON_ReplaceItemInArray(cJSON *array, int which, cJSON *newitem);
00245 cJSON_PUBLIC(cJSON_bool) cJSON_ReplaceItemInObject(cJSON *object, const char *string, cJSON *newitem);
00246 cJSON_PUBLIC(cJSON_bool) cJSON_ReplaceItemInObjectCaseSensitive(cJSON *object, const char *string, cJSON

```

```

        *newitem);
00247
00248 /* Duplicate a cJSON item */
00249 cJSON_PUBLIC(cJSON *) cJSON_Duplicate(const cJSON *item, cJSON_bool recurse);
00250 /* Duplicate will create a new, identical cJSON item to the one you pass, in new memory that will
00251  * need to be released. With recurse!=0, it will duplicate any children connected to the item.
00252  * The item->next and ->prev pointers are always zero on return from Duplicate. */
00253 /* Recursively compare two cJSON items for equality. If either a or b is NULL or invalid, they will be
    considered unequal.
00254  * case_sensitive determines if object keys are treated case sensitive (1) or case insensitive (0) */
00255 cJSON_PUBLIC(cJSON_bool) cJSON_Compare(const cJSON * const a, const cJSON * const b, const cJSON_bool
    case_sensitive);
00256
00257 /* Minify a strings, remove blank characters(such as ' ', '\t', '\r', '\n') from strings.
00258  * The input pointer json cannot point to a read-only address area, such as a string constant,
00259  * but should point to a readable and writable address area. */
00260 cJSON_PUBLIC(void) cJSON_Minify(char *json);
00261
00262 /* Helper functions for creating and adding items to an object at the same time.
00263  * They return the added item or NULL on failure. */
00264 cJSON_PUBLIC(cJSON*) cJSON_AddNullToObject(cJSON * const object, const char * const name);
00265 cJSON_PUBLIC(cJSON*) cJSON_AddTrueToObject(cJSON * const object, const char * const name);
00266 cJSON_PUBLIC(cJSON*) cJSON_AddFalseToObject(cJSON * const object, const char * const name);
00267 cJSON_PUBLIC(cJSON*) cJSON_AddBoolToObject(cJSON * const object, const char * const name, const
    cJSON_bool boolean);
00268 cJSON_PUBLIC(cJSON*) cJSON_AddNumberToObject(cJSON * const object, const char * const name, const
    double number);
00269 cJSON_PUBLIC(cJSON*) cJSON_AddStringToObject(cJSON * const object, const char * const name, const char
    * const string);
00270 cJSON_PUBLIC(cJSON*) cJSON_AddRawToObject(cJSON * const object, const char * const name, const char *
    const raw);
00271 cJSON_PUBLIC(cJSON*) cJSON_AddObjectToObject(cJSON * const object, const char * const name);
00272 cJSON_PUBLIC(cJSON*) cJSON_AddArrayToObject(cJSON * const object, const char * const name);
00273
00274 /* When assigning an integer value, it needs to be propagated to valuedouble too. */
00275 #define cJSON_SetIntValue(object, number) ((object) ? (object)->valueint = (object)->valuedouble =
    (number) : (number))
00276 /* helper for the cJSON_SetNumberValue macro */
00277 cJSON_PUBLIC(double) cJSON_SetNumberHelper(cJSON *object, double number);
00278 #define cJSON_SetNumberValue(object, number) ((object != NULL) ? cJSON_SetNumberHelper(object,
    (double)number) : (number))
00279 /* Change the valuelstring of a cJSON_String object, only takes effect when type of object is
    cJSON_String */
00280 cJSON_PUBLIC(char*) cJSON_SetValuelstring(cJSON *object, const char *valuelstring);
00281
00282 /* If the object is not a boolean type this does nothing and returns cJSON_Invalid else it returns the
    new type*/
00283 #define cJSON_SetBoolValue(object, boolValue) ( \
00284     (object != NULL && ((object)->type & (cJSON_False|cJSON_True))) ? \
00285     (object)->type=((object)->type & ~(cJSON_False|cJSON_True)) | ((boolValue)?cJSON_True:cJSON_False)
    : \
00286     cJSON_Invalid\
00287 )
00288
00289 /* Macro for iterating over an array or object */
00290 #define cJSON_ArrayForEach(element, array) for(element = (array != NULL) ? (array)->child : NULL;
    element != NULL; element = element->next)
00291
00292 /* malloc/free objects using the malloc/free functions that have been set with cJSON_InitHooks */
00293 cJSON_PUBLIC(void *) cJSON_malloc(size_t size);
00294 cJSON_PUBLIC(void) cJSON_free(void *object);
00295
00296 #ifdef __cplusplus
00297 }
00298 #endif
00299
00300 #endif

```

4.3 cJSON_Utils.h

```

00001 /*
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00003
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```

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00020 THE SOFTWARE.
00021 */
00022
00023 #ifndef cJSON_Utils__h
00024 #define cJSON_Utils__h
00025
00026 #ifdef __cplusplus
00027 extern "C"
00028 {
00029 #endif
00030
00031 #include "cJSON.h"
00032
00033 /* Implement RFC6901 (https://tools.ietf.org/html/rfc6901) JSON Pointer spec. */
00034 cJSON_PUBLIC(cJSON *) cJSONUtils_GetPointer(cJSON * const object, const char *pointer);
00035 cJSON_PUBLIC(cJSON *) cJSONUtils_GetPointerCaseSensitive(cJSON * const object, const char *pointer);
00036
00037 /* Implement RFC6902 (https://tools.ietf.org/html/rfc6902) JSON Patch spec. */
00038 /* NOTE: This modifies objects in 'from' and 'to' by sorting the elements by their key */
00039 cJSON_PUBLIC(cJSON *) cJSONUtils_GeneratePatches(cJSON * const from, cJSON * const to);
00040 cJSON_PUBLIC(cJSON *) cJSONUtils_GeneratePatchesCaseSensitive(cJSON * const from, cJSON * const to);
00041 /* Utility for generating patch array entries. */
00042 cJSON_PUBLIC(void) cJSONUtils_AddPatchToArray(cJSON * const array, const char * const operation, const
char * const path, const cJSON * const value);
00043 /* Returns 0 for success. */
00044 cJSON_PUBLIC(int) cJSONUtils_ApplyPatches(cJSON * const object, const cJSON * const patches);
00045 cJSON_PUBLIC(int) cJSONUtils_ApplyPatchesCaseSensitive(cJSON * const object, const cJSON * const
patches);
00046
00047 /*
00048 // Note that ApplyPatches is NOT atomic on failure. To implement an atomic ApplyPatches, use:
00049 //int cJSONUtils_AtomicApplyPatches(cJSON **object, cJSON *patches)
00050 //{
00051 //    cJSON *modme = cJSON_Duplicate(*object, 1);
00052 //    int error = cJSONUtils_ApplyPatches(modme, patches);
00053 //    if (!error)
00054 //    {
00055 //        cJSON_Delete(*object);
00056 //        *object = modme;
00057 //    }
00058 //    else
00059 //    {
00060 //        cJSON_Delete(modme);
00061 //    }
00062 //    return error;
00063 //}
00064 // Code not added to library since this strategy is a LOT slower.
00065 */
00066
00067 /* Implement RFC7386 (https://tools.ietf.org/html/rfc7386) JSON Merge Patch spec. */
00068 /* target will be modified by patch. return value is new ptr for target. */
00069 cJSON_PUBLIC(cJSON *) cJSONUtils_MergePatch(cJSON *target, const cJSON * const patch);
00070 cJSON_PUBLIC(cJSON *) cJSONUtils_MergePatchCaseSensitive(cJSON *target, const cJSON * const patch);
00071 /* generates a patch to move from -> to */
00072 /* NOTE: This modifies objects in 'from' and 'to' by sorting the elements by their key */
00073 cJSON_PUBLIC(cJSON *) cJSONUtils_GenerateMergePatch(cJSON * const from, cJSON * const to);
00074 cJSON_PUBLIC(cJSON *) cJSONUtils_GenerateMergePatchCaseSensitive(cJSON * const from, cJSON * const
to);
00075
00076 /* Given a root object and a target object, construct a pointer from one to the other. */
00077 cJSON_PUBLIC(char *) cJSONUtils_FindPointerFromObjectTo(const cJSON * const object, const cJSON *
const target);
00078
00079 /* Sorts the members of the object into alphabetical order. */
00080 cJSON_PUBLIC(void) cJSONUtils_SortObject(cJSON * const object);
00081 cJSON_PUBLIC(void) cJSONUtils_SortObjectCaseSensitive(cJSON * const object);
00082
00083 #ifdef __cplusplus
00084 }
00085 #endif
00086 #endif
00087
00088 #endif

```

4.4 daq.h

```

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00020  */
00021
00022 /*
00023  * REMOVE THE CHIP PROGRAMMER, IT CHANGES THE ADC READINGS
00024  */
00025 /*
00026  * 12-bit analog 64 sample average per channel on ports A and B
00027  * uses the compute burst average mode threshold interrupt to auto change
00028  * channels during interrupt after a repeat count
00029  */
00030
00031 // This is a guard condition so that contents of this file are not included
00032 // more than once.
00033 #ifndef DAQ_H
00034 #define DAQ_H
00035
00036 #include <math.h>
00037 #include "mcc_generated_files/spi1.h"
00038 #include "mcc_generated_files/crc.h"
00039 #include "vconfig.h"
00040 #include "timers.h"
00041 #include "tests.h"
00042
00043 extern struct V_data V;
00044 /*
00045  * adc channel configuration
00046  */
00047 #define LAST_ADC_CHAN 0XD
00048 #define ADC_BUFFER_SIZE 16 // MUST BE AT LEAST 1
00049 #define ADC_SCAN_SPEED 200 // sample timer speed in ms
00050 #define ADC_SCAN_CHAN 0b0011110001110111 // convert these analog ports bitmap
00051
00052 /*
00053  * Normal or high volt channels (bit set)
00054  */
00055 #define ADC_V_CHAN_TYPE 0b0010000000100000
00056
00057 /*
00058  * current sensor configuration
00059  */
00060 // #define BAT_100A // USE the 100A honeywell sensor for battery current, reduces range but has better
// data within that range
00061 #define NUM_C_SENSORS 3
00062 #define ADC_C_CHAN 0b0000100000000011 // 5 volt hall current sensor bitmap
00063 #define ADC_C_CHAN_TYPE 0b0000000000000001 // 0 - 100A, 1=200A or battery 100A type if BAT_100A is
// defined
00064 #define ADC_C_CHAN_MPPT 0b0000100000000000 // 0 - 100A mppt channel
00065 /*
00066  * temp sensor configuration
00067  */
00068 #define NUM_T_SENSORS 0
00069 #define ADC_T_CHAN 0b0000000000000000 // 5 volt temp sensor bitmap
00070 #define ADC_T_CHAN_TYPE 0b0000000000000000 // type bitmap
00071
00072 /*
00073  * dac channel configuration
00074  */
00075 #define LAST_DAC_CHAN 0X1
00076 #define DAC_BUFFER_SIZE 2 // MUST BE AT LEAST 1
00077 #define DAC_LOAD_A 0b0100
00078 #define DAC_LOAD_B 0b0101
00079 #define DCHAN_A 0
00080 #define DCHAN_B 1
00081 #define DAC_SCALE 0.002442f // 10 vdc @ 12-bits resolution per bit
00082
00083 #define ZERO_RANGE 100
00084 #define TEN_A_RANGE 100
00085
00086 #define EE_CHECKMARK 0x1957

```

```

00087 #define TATE          0x42
00088
00089 /*
00090  * current sensor configuration
00091  */
00092 //#define CAL_10A    // set 10A scaling
00093 //#define CAL_TWO // dupe the PV sensor results into the MPPT calibration table
00094
00095 /*
00096  * conversion constants
00097  */
00098 /*
00099  * REMOVE THE CHIP PROGRAMMER, IT CHANGES THE ADC READINGS
00100  */
00101 /*
00102  * 13-bit adc result ADRPT 64 samples and a ADCRS of 5 [0..8190] steps
00103  */
00104 #define C_SCALE      0.6250000f
00105 #define V_SCALE      4.1254125f
00106 #define V_SCALE_H    8.2000000f
00107
00108 #ifndef BAT_100A
00109 #define A200          0 // BATTERY,  adc line 0
00110 #endif
00111 #define A100B          0 // BATTERY,  adc line 0
00112 #define A100           1 // CV,      adc line 1
00113 #define A100M          2 // PV,      adc line 1
00114
00115 #ifndef BAT_100A
00116 #define C_A200          0.0361010f // BATTERY Amp scalar, second line [0], 0.0862000f, BO PLUG
00117 #endif
00118 #define C_A100B          0.0361010f // PV Amp scalar, second line [0]
00119 #define C_A100           0.0361010f // PV Amp scalar, first line [1]
00120 #define C_A100M          0.0862000f // MPPT Amp scalar, first line [2]
00121
00122 #ifndef BAT_100A
00123 #define C_OFFSET200 3965 // BATTERY sensor adc zero Amp counts, line 0,  BO sensor PLUG, orig value
3965
00124 #endif
00125 #define C_OFFSET100B 3934 // BATTERY sensor adc zero Amp counts, line 0
00126 #define C_OFFSET100 3995 // PV sensor adc zero Amp counts, line 1
00127 #define C_OFFSET100M 3939 // MPPT sensor adc zero Amp counts, line 2, Orig value 3934, adjust
this for DL bat current zero
00128
00129 /*
00130  * sanity check values for current calibration routines
00131  */
00132 #define C_CAL_ZERO 4000
00133 #ifndef BAT_100A
00134 #define C_CAL_A200 4150 // fixme with real values
00135 #endif
00136 #define C_CAL_A100 4240 // ..
00137
00138 #include <xc.h> // include processor files - each processor file is guarded.
00139 #include "mcc_generated_files/adcc.h"
00140 #include "mcc_generated_files/pin_manager.h"
00141 #include "mcc_generated_files/memory.h"
00142
00143 typedef enum {
00144     CONV, // auto system conversion
00145     Q_CONV, // convert to voltage for calibration/offsets
00146 } adc_conv_t;
00147
00148 bool start_adc_scan(void);
00149 bool check_adc_scan(void);
00150 void clear_adc_scan(void);
00151 bool update_adc_result(void);
00152 adc_result_t get_raw_result(const adcc_channel_t);
00153 float conv_raw_result(const adcc_channel_t, const adc_conv_t);
00154 void dac_spi_control(bool);
00155 void set_dac(void);
00156 uint16_t set_dac_a(float);
00157 uint16_t set_dac_b(float);
00158 bool cal_current_zero(bool, int16_t, int16_t, int16_t);
00159 bool cal_current_10A(bool, int16_t, int16_t, float, float);
00160 bool read_cal_data(void);
00161 void write_cal_data(void);
00162 void update_cal_data(void);
00163 bool update_hist_data(bool, volatile hist_type*);
00164 void set_hist_flag(void);
00165 #endif
00166

```


4.5 dio.h

```

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00020 */
00021
00022 /*
00023 * File:
00024 * Author:
00025 * Comments:
00026 * Revision history:
00027 */
00028
00029 // This is a guard condition so that contents of this file are not included
00030 // more than once.
00031 #ifndef DIO_H
00032 #define DIO_H
00033
00034 #include <xc.h> // include processor files - each processor file is guarded.
00035 #include "vconfig.h"
00036 #include "tests.h"
00037
00038 #define NUM_SWITCHES 6
00039
00040 typedef struct rbutton_type {
00041     SW_STATES sw;
00042     uint32_t count;
00043 } rbutton_type;
00044
00045 void start_switch_handler(void);
00046 SW_STATES get_switch(const uint8_t);
00047 SW_STATES get_switch_bm(const uint8_t);
00048 rbutton_type get_switch_data(uint8_t);
00049 uint8_t check_switches(void);
00050 void clear_switch(uint8_t);
00051
00052 #endif /* DIO_H */
00053

```

4.6 eadog.h

```

00001 /*
00002 * File: eadog.h
00003 * Author: root
00004 *
00005 * Created on July 27, 2015, 2:05 PM
00006 */
00007
00008 /* Parts taken from:
00009 * Modified for C18 and PIC18F45K80 with SPI using ring buffers with interrupts
00010 *
00011 * file: EA-DOGM_MIO.c
00012 * version: 2.03
00013 * description: Multi I/O driver for EA DOGM displays
00014 * : Uses 8Bit, SPI HW or SPI SW (bitbang)
00015 * written by : Michael Bradley (mbradley@mcu-labs.com)
00016 * contributions: Imaginos (CCS forum), Emil Nad (8Bit testing)
00017 * jgschmidt (CCS forum)
00018 */
00019
00020 #ifndef EADOG_H
00021 #define EADOG_H
00022
00023 #ifdef __cplusplus
00024 extern "C" {

```



```

00025 #endif
00026 #include "vconfig.h"
00027 #include "ringbufs.h"
00028
00029     extern struct V_data V;
00030
00031 // #define USEMCC_SPI
00032
00033 #define USE_LCD_DMA
00034
00035 #define NSB      5
00036 #define LSB      21
00037
00038 #define LCD_PWR_DELAY    350000
00039
00040 #define LCD_CMD_MASK     0x01
00041 #define LCD_CMD_SET      0x100
00042 #define LCD_CLEAR_HOME   0x04
00043 #define NHD_CMD          0xFE
00044 #define NHD_BL_OFF       1
00045 #define NHD_BL_LOW       2
00046 #define NHD_BL_MED       5
00047 #define NHD_BL_HIGH      8
00048
00049 #define LCD_CMD_MASK     0x01
00050 #define LCD_CMD_SET      0x100
00051 #define LCD_CLEAR_HOME   0x04
00052 #define LCD_CMD_ON       0x41
00053 #define LCD_CMD_OFF      0x42
00054 #define LCD_CMD_HOME     0x46
00055 #define LCD_CMD_CLR      0x51
00056 #define LCD_CMD_CONT     0x52
00057 #define LCD_CMD_BRI      0x53
00058 #define NHD_CMD          0xFE
00059 #define NHD_POS          0x45
00060 #define NHD_BL_OFF       1
00061 #define NHD_BL_LOW       2
00062 #define NHD_BL_MED       5
00063 #define NHD_BL_HIGH      8
00064 #define NHD_CONT         45
00065
00066 #define NHD_T_DELAY      8
00067 #define NHD_S_DELAY      200
00068 #define NHD_L_DELAY      800
00069
00070 #define LCD0             0
00071 #define LCD1             1
00072 #define LCD2             2
00073 #define LCD3             3
00074
00075     void wdtdelay(uint32_t);
00076     void init_display(void);
00077     void init_port_dma(void);
00078     void send_port_data_dma(uint16_t);
00079     uint8_t* port_data_dma_ptr(void);
00080     void send_lcd_data_dma(uint8_t);
00081     void send_lcd_cmd_dma(uint8_t);
00082     void start_lcd(void);
00083     void wait_lcd_set(void);
00084     bool wait_lcd_check(void);
00085     void wait_lcd_done(void);
00086     void eaDogM_WriteChr(int8_t);
00087     void eaDogM_WriteCommand(uint8_t);
00088     void eaDogM_SetPos(uint8_t, uint8_t);
00089     void eaDogM_ClearRow(uint8_t);
00090     void eaDogM_WriteString(char *);
00091     void eaDogM_WriteStringAtPos(uint8_t, uint8_t, char *);
00092     void eaDogM_WriteIntAtPos(uint8_t, uint8_t, uint8_t);
00093     void eaDogM_WriteByteToCGRAM(uint8_t, uint8_t);
00094
00095     void clear_lcd_done(void);
00096     void spi_rec_done(void);
00097
00098 #define eaDogM_Cls()          eaDogM_WriteCommand(EADOGM_CMD_CLR)
00099 #define eaDogM_CursorOn()    eaDogM_WriteCommand(EADOGM_CMD_CURSOR_ON)
00100 #define eaDogM_CursorOff()   eaDogM_WriteCommand(EADOGM_CMD_CURSOR_OFF)
00101 #define eaDogM_DisplayOn()   eaDogM_WriteCommand(EADOGM_CMD_DISPLAY_ON)
00102 #define eaDogM_DisplayOff()  eaDogM_WriteCommand(EADOGM_CMD_DISPLAY_OFF)
00103
00104 #ifdef __cplusplus
00105 }
00106 #endif
00107
00108 #endif /* EADOG_H */
00109

```

4.7 gemsecs.h

```

00001 /*
00002  * File:   gemsecs.h
00003  * Author: root
00004  *
00005  * Created on February 26, 2019, 2:58 PM
00006  */
00007
00008 #ifndef GEMSECS_H
00009 #define GEMSECS_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #define ENQ 0x05
00016 #define EOT 0x04
00017 #define ACK 0x06
00018 #define NAK 0x15
00019
00020 #include <stdio.h>
00021 #include <string.h>
00022 #include "vconfig.h"
00023 #include "mcc_generated_files/mcc.h"
00024 #include "mcc_generated_files/uart1.h"
00025 #include "timers.h"
00026 #include "mydisplay.h"
00027 #include "msg_text.h"
00028
00029     typedef struct block10_type {
00030         uint32_t systemb;
00031         uint8_t bidl;
00032         uint8_t bidh : 7;
00033         uint8_t ebit : 1;
00034         uint8_t function;
00035         uint8_t stream : 7;
00036         uint8_t wbit : 1;
00037         uint8_t didl;
00038         uint8_t didh : 7;
00039         uint8_t rbit : 1;
00040     } block10_type;
00041
00042     typedef union block10 {
00043         uint8_t b[sizeof(block10_type)];
00044         block10_type block;
00045     } block10;
00046
00047     typedef struct header10 {
00048         uint16_t checksum;
00049         block10 block;
00050         uint8_t length;
00051     } header10;
00052
00053     typedef struct header12 {
00054         uint16_t checksum;
00055         uint8_t data[2];
00056         block10 block;
00057         uint8_t length;
00058     } header12;
00059
00060     typedef struct header13 {
00061         uint16_t checksum;
00062         uint8_t data[3];
00063         block10 block;
00064         uint8_t length;
00065     } header13;
00066
00067     typedef struct header14 {
00068         uint16_t checksum;
00069         uint8_t data[4];
00070         block10 block;
00071         uint8_t length;
00072     } header14;
00073
00074     typedef struct header17 {
00075         uint16_t checksum;
00076         uint8_t data[7];
00077         block10 block;
00078         uint8_t length;
00079     } header17;
00080
00081     typedef struct header18 {
00082         uint16_t checksum;
00083         uint8_t data[8];
00084         block10 block;

```

```

00085     uint8_t length;
00086 } header18;
00087
00088 typedef struct header24 {
00089     uint16_t checksum;
00090     uint8_t data[12];
00091     uint8_t datam[2];
00092     block10 block;
00093     uint8_t length;
00094 } header24;
00095
00096 typedef struct header26 {
00097     uint16_t checksum;
00098     uint8_t data[14];
00099     uint8_t datam[2];
00100     block10 block;
00101     uint8_t length;
00102 } header26;
00103
00104 typedef struct header27 {
00105     uint16_t checksum;
00106     uint8_t data[17];
00107     block10 block;
00108     uint8_t length;
00109 } header27;
00110
00111 typedef struct header33 {
00112     uint16_t checksum;
00113     uint8_t data[23];
00114     block10 block;
00115     uint8_t length;
00116 } header33;
00117
00118 typedef struct header153 {
00119     uint16_t checksum;
00120     uint8_t data[143];
00121     block10 block;
00122     uint8_t length;
00123 } header153;
00124
00125 typedef struct header254 {
00126     uint16_t checksum;
00127     uint8_t data[244];
00128     block10 block;
00129     uint8_t length;
00130 } header254;
00131
00132 typedef struct response_type {
00133     uint8_t *header;
00134     uint8_t length;
00135     uint8_t *reply;
00136     uint8_t reply_length;
00137     uint8_t respond : 1;
00138 } response_type;
00139
00140 typedef struct gem_message_type {
00141     header33 message;
00142     response_type block;
00143     uint16_t delay;
00144     uint8_t stack;
00145 } gem_message_type;
00146
00147 typedef struct gem_display_type {
00148     header153 message;
00149     response_type block;
00150     uint16_t delay;
00151     uint8_t stack;
00152 } gem_display_type;
00153
00154 uint16_t block_checksum(uint8_t *, const uint16_t);
00155 uint16_t run_checksum(const uint8_t, const bool);
00156 LINK_STATES m_protocol(LINK_STATES *);
00157 LINK_STATES r_protocol(LINK_STATES *);
00158 LINK_STATES t_protocol(LINK_STATES *);
00159 void hb_message(void);
00160 void terminal_format(DISPLAY_TYPES);
00161 uint16_t format_display_text(const char *);
00162 P_CODES s10f11_opcmd(void);
00163 uint16_t s6f11_opcmd(void);
00164 response_type secs_II_message(uint8_t, uint8_t);
00165 bool sequence_messages(uint8_t);
00166 bool gem_messages(response_type *, uint8_t);
00167 void secs_II_monitor_message(uint8_t, uint8_t, uint16_t);
00168 GEM_STATES secs_gem_state(uint8_t, uint8_t);
00169
00170 #ifdef __cplusplus
00171 }

```

```

00172 #endif
00173
00174 #endif /* GEMSECS_H */
00175

```

4.8 hid.h

```

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00020 */
00021
00022 /*
00023 * File:
00024 * Author:
00025 * Comments:
00026 * Revision history:
00027 */
00028
00029 // This is a guard condition so that contents of this file are not included
00030 // more than once.
00031 #ifndef HID_H
00032 #define HID_H
00033
00034 #include <xc.h> // include processor files - each processor file is guarded.
00035 #include "dio.h"
00036
00037 typedef enum {
00038     HID_MAIN = 0,
00039     HID_PWR,
00040     HID_RUN,
00041     HID_AUX,
00042     HID_LAST,
00043 } H_CODES;
00044
00045 typedef enum {
00046     H_STATE_INIT = 0,
00047     H_STATE_DISPLAY,
00048     H_STATE_WAIT,
00049     H_STATE_LOG,
00050     H_STATE_COMM,
00051     H_STATE_ERROR
00052 } H_STATES;
00053
00054 typedef struct H_data { // control data structure
00055     H_CODES hid_display;
00056     H_STATES h_state;
00057     bool wait_select, select_p, wait_enter, enter_p, wait_calib, calib_p; // buttons pressed should
00058     clean      uint8_t sequence;
00059 } H_data;
00060
00061 H_data* hid_input(H_data*);
00062 H_data* hid_display(H_data*);
00063 bool check_enter_button(H_data*);
00064 bool check_calib_button(H_data*);
00065 void clear_hid_pflags(H_data*);
00066 void clear_hid_wflags(H_data*);
00067
00068 #endif /* HID_H */
00069

```

4.9 mbmc.h

```

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00021
00022 /*
00023 * File:
00024 * Author:
00025 * Comments:
00026 * Revision history:
00027 */
00028
00029 // This is a guard condition so that contents of this file are not included
00030 // more than once.
00031 #ifndef MBMC_H
00032 #define MBMC_H
00033
00034 #include <xc.h> // include processor files - each processor file is guarded.
00035 #include <time.h>
00036 #include "vconfig.h"
00037 #include "daq.h"
00038
00039 extern V_data V;
00040
00041 #define LOCALTIME_OFFSET 60ul*60ul*(-8ul)
00042 #define LP_BUFFER_SIZE ADC_BUFFER_SIZE+8
00043 #define STATIC_LOAD_POWER 14.00f
00044 #define BANKAH 200.0f // baatery bank label Ah
00045 #define DEGRADE 25.0f // adjust to what the battery bank Ah really is
00046 #define SSLICE 3600.0f // samples per hour
00047
00048 #ifdef DEBUG_HIST1
00049 #define DUPDATE 600 // after dark update time offset
00050 #else
00051 #define DUPDATE 14400 // after dark update time offset
00052 #endif
00053
00054 #define BLOAD1 10.0f // load resistor in ohms
00055 #define BLOAD2 2.5f // ditto
00056 #define MAX_LOAD_TIME 75
00057
00058 #define ROR_LIMIT_LOW 0.0140f
00059 #define ROR_LIMIT_SET 0.0001f
00060 #define ROR_LIMIT_NOISE 0.0051f
00061
00062 #define ROR_WAIT 20
00063 #define ROR_TIMES 1
00064
00065 #define DAWN_VOLTS 22.0f
00066 #define DUSK_VOLTS 12.0f
00067 #define BRIGHT_VOLTS 50.0f
00068 #define BAT_LOW_VOLTS 21.5f
00069 #define BAT_DEAD_VOLTS 18.0f
00070 #define MAX_AC_TIME 60*60*4 // AC charger max runtime per trigger
00071 #define DAY_DELAY 240
00072 #define SOC_TOO_LOW 20
00073 #define SOC_CRITICAL 3
00074
00075 #define CEF 1.0f
00076 #define PEUKERT 1.0f
00077
00078 typedef struct C_data { // real calculated variables
00079     float calc[ADC_BUFFER_SIZE];
00080     float c_load, c_bat, c_pv, c_mppt, v_cc, v_pv, v_bat, v_cbus, v_bbat, v_temp, v_inverter,
00081     v_sensor, v_cmode, bv_ror, bc_ror;
00082     float p_load, p_inverter, p_pv, p_bat, p_mppt, start_power;
00083     float t_comp, esr;
00084     float bank_ah, dynamic_ah, pv_ah, loadah, dynamic_ah_adj, dynamic_ah_daily, dynamic_ah_adj_daily;

```

```

00084     float bkwi, bkwo, pvkw, invkw, loadkw;
00085     uint16_t runtime, soc;
00086     bool update;
00087     hist_type hist[1];
00088     float load_i1, load_i2, bv_noload, bv_one_load, bv_full_load;
00089     bool day, dupdate;
00090     time_t day_start, day_end, day_update, updates;
00091 } C_data;
00092
00093 typedef enum {
00094     WIDE_ZERO = 1,
00095     NO_NEG = 2,
00096     WIDE_NEG = 4,
00097     SKIP = 8,
00098 } FIX_CODES;
00099
00100 extern volatile C_data C;
00101
00102 float lp_filter(const float, const uint8_t, const int8_t);
00103 void convert_adc_data(void);
00104 void calc_model_data(void);
00105 void calc_ror_data(void);
00106 void static_soc(void);
00107 void set_load_relay_zero(const bool);
00108 void set_load_relay_one(const bool);
00109 void set_load_relay_two(const bool);
00110 void set_ac_charger_relay(const bool);
00111 bool get_ac_charger_relay(void);
00112 bool check_day_time(void);
00113 void load_hist_data(void);
00114 float calc_fixups(float data, FIX_CODES fixup);
00115
00116 char spinners(uint8_t, uint8_t);
00117 time_t time(time_t *);
00118 void set_time(const time_t t);
00119 #endif

```

4.10 adcc.h

```

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00046 */
00047 #ifndef ADCC_H
00048 #define ADCC_H
00049
00053
00054 #include <xc.h>
00055 #include <stdint.h>
00056 #include <stdbool.h>
00057
00058 #ifdef __cplusplus // Provide C++ Compatibility
00059     extern "C" {
00060
00061
00062 #endif
00063
00067
00071
00072 typedef uint16_t adc_result_t;

```

```
00073 #ifndef uint24_t
00074 typedef __uint24 uint24_t;
00075 #endif
00076
00088
00089 typedef enum
00090 {
00091     C_BATT = 0x0,
00092     C_PV = 0x1,
00093     V_CC = 0x2,
00094     V_BAT = 0x4,
00095     V_PV = 0x5,
00096     V_CMODE = 0x6,
00097     V_BBAT = 0xA,
00098     C_MPPT = 0xB,
00099     V_INVERTER = 0xC,
00100     V_LIGHT_SENSOR = 0xD,
00101     channel_VSS = 0x3B,
00102     channel_Temp = 0x3C,
00103     channel_DAC1 = 0x3D,
00104     channel_FVR_Buffer1 = 0x3E,
00105     channel_FVR_Buffer2 = 0x3F
00106 } adcc_channel_t;
00107
00111
00140 void ADCC_Initialize(void);
00141
00169 void ADCC_StartConversion(adcc_channel_t channel);
00170
00199 bool ADCC_IsConversionDone(void);
00200
00231 adc_result_t ADCC_GetConversionResult(void);
00232
00262 adc_result_t ADCC_GetSingleConversion(adcc_channel_t channel);
00263
00287 void ADCC_StopConversion(void);
00288
00314 void ADCC_SetStopOnInterrupt(void);
00315
00339 void ADCC_DischargeSampleCapacitor(void);
00340
00365 void ADCC_LoadAcquisitionRegister(uint16_t);
00366
00391 void ADCC_SetPrechargeTime(uint16_t);
00392
00416 void ADCC_SetRepeatCount(uint8_t);
00417
00444 uint8_t ADCC_GetCurrentCountofConversions(void);
00445
00468 void ADCC_ClearAccumulator(void);
00469
00493 uint24_t ADCC_GetAccumulatorValue(void);
00494
00521 bool ADCC_HasAccumulatorOverflowed(void);
00522
00546 uint16_t ADCC_GetFilterValue(void);
00547
00574 uint16_t ADCC_GetPreviousResult(void);
00575
00600 void ADCC_DefineSetPoint(uint16_t);
00601
00626 void ADCC_SetUpperThreshold(uint16_t);
00627
00652 void ADCC_SetLowerThreshold(uint16_t);
00653
00679 uint16_t ADCC_GetErrorCalculation(void);
00680
00706 void ADCC_EnableDoubleSampling(void);
00707
00730 void ADCC_EnableContinuousConversion(void);
00731
00754 void ADCC_DisableContinuousConversion(void);
00755
00782 bool ADCC_HasErrorCrossedUpperThreshold(void);
00783
00810 bool ADCC_HasErrorCrossedLowerThreshold(void);
00811
00837 uint8_t ADCC_GetConversionStageStatus(void);
00838
00839
00840
00854 void ADCC_SetADIInterruptHandler(void (* InterruptHandler)(void));
00855
00856
00870 void ADCC_SetADTIIInterruptHandler(void (* InterruptHandler)(void));
00871
00872
```

```

00889 void ADCC_DefaultInterruptHandler(void);
00890 #ifdef __cplusplus // Provide C++ Compatibility
00891     }
00892 }
00893 #endif
00894 #endif
00895 //ADCC_H
00896 #endif
00900

```

4.11 clkref.h

```

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00046 */
00047 #ifndef CLKREF_H
00048 #define CLKREF_H
00049
00053
00054 #include <stdbool.h>
00055 #include <stdint.h>
00056
00057 #ifdef __cplusplus // Provide C++ Compatibility
00058     extern "C" {
00059     #endif
00060
00061 #endif
00062
00066
00092 void CLKREF_Initialize(void);
00093
00094 #ifdef __cplusplus // Provide C++ Compatibility
00095     }
00096 }
00097 #endif
00098 #endif
00099 // CLKREF_H
00100 #endif
00104
00105

```

4.12 crc.h

```

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```



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00045 */
00046
00047 #ifndef CRC_H
00048 #define CRC_H
00049
00053
00054 #include <xc.h>
00055 #include <stdint.h>
00056 #include <stdbool.h>
00057
00058 // Provide C++ compatability
00059 #ifdef __cplusplus
00060
00061 extern "C" {
00062
00063 #endif
00064
00068
00069 #define NORMAL 0
00070 #define REVERSE 1
00071
00075
00123 void CRC_Initialize(void);
00124
00164 void CRC_Start(void);
00165
00203 bool CRC_8BitDataWrite(uint8_t data);
00204
00241 uint16_t CRC_CalculatedResultGet(bool reverse, uint16_t xorValue);
00242
00280 bool CRC_IsBusy(void);
00281
00320 void CRC_SCAN_StartScanner(void);
00321
00360 void CRC_SCAN_StopScanner(void);
00361
00401 void CRC_SCAN_SetAddressLimit(uint32_t startAddr, uint32_t endAddr);
00402
00445 bool CRC_SCAN_IsScannerBusy(void);
00446
00486 bool CRC_SCAN_HasScanCompleted(void);
00487
00488
00489
00490 #ifdef __cplusplus
00491 }
00492 #endif
00493
00494 #endif /* _CRC_H */

```

4.13 device_config.h

```

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00045 */
00046
00047 #ifndef DEVICE_CONFIG_H
00048 #define DEVICE_CONFIG_H
00049
00050 #define _XTAL_FREQ 64000000
00051
00052 #endif /* DEVICE_CONFIG_H */

```

4.14 dma1.h

```

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00044     SOFTWARE.
00045 */
00046
00047 #ifndef DMA1_H
00048 #define DMA1_H
00049
00053 #include <stdint.h>
00054
00055 uint8_t lcd_dma_buf[32];
00056
00063 void DMA1_Initialize(void);
00064
00070 void DMA1_SelectSourceRegion(uint8_t region);
00071
00077 void DMA1_SetSourceAddress(uint24_t address);
00078
00084 void DMA1_SetDestinationAddress(uint16_t address);
00085
00091 void DMA1_SetSourceSize(uint16_t size);
00092
00098 void DMA1_SetDestinationSize(uint16_t size);
00099
00105 uint24_t DMA1_GetSourcePointer(void);
00106
00112 uint16_t DMA1_GetDestinationPointer(void);
00113
00119 void DMA1_SetStartTrigger(uint8_t sirq);
00120
00126 void DMA1_SetAbortTrigger(uint8_t airq);
00127
00133 void DMA1_StartTransfer(void);
00134
00140 void DMA1_StartTransferWithTrigger(void);
00141
00147 void DMA1_StopTransfer(void);
00148
00154 void DMA1_SetDMAPriority(uint8_t priority);
00155
00161 void DMA1_SetSCNTIInterruptHandler(void (* InterruptHandler)(void));
00162

```

```

00168 void DMA1_SetDCNTIInterruptHandler(void (* InterruptHandler)(void));
00169
00175 void DMA1_SetAIInterruptHandler(void (* InterruptHandler)(void));
00176
00182 void DMA1_SetORIInterruptHandler(void (* InterruptHandler)(void));
00188 void DMA1_DefaultInterruptHandler(void);
00189 #endif //DMA1_H

```

4.15 dma2.h

```

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00045 */
00046
00047 #ifndef DMA2_H
00048 #define DMA2_H
00049
00053 #include <stdint.h>
00054
00055 uint8_t uart1_dma_buf[32];
00056
00063 void DMA2_Initialize(void);
00064
00070 void DMA2_SelectSourceRegion(uint8_t region);
00071
00077 void DMA2_SetSourceAddress(uint24_t address);
00078
00084 void DMA2_SetDestinationAddress(uint16_t address);
00085
00091 void DMA2_SetSourceSize(uint16_t size);
00092
00098 void DMA2_SetDestinationSize(uint16_t size);
00099
00105 uint24_t DMA2_GetSourcePointer(void);
00106
00112 uint16_t DMA2_GetDestinationPointer(void);
00113
00119 void DMA2_SetStartTrigger(uint8_t sirq);
00120
00126 void DMA2_SetAbortTrigger(uint8_t airq);
00127
00133 void DMA2_StartTransfer(void);
00134
00140 void DMA2_StartTransferWithTrigger(void);
00141
00147 void DMA2_StopTransfer(void);
00148
00154 void DMA2_SetDMAPriority(uint8_t priority);
00155
00156
00157
00158
00159 #endif //DMA2_H

```

4.16 ext_int.h

```

00001

```

```

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00045  */
00046
00047  #ifndef _EXT_INT_H
00048  #define _EXT_INT_H
00049
00053  #include <xc.h>
00054
00055  // Provide C++ Compatibility
00056  #ifdef __cplusplus
00057
00058  extern "C" {
00059
00060  #endif
00061
00093  #define EXT_INT0_InterruptFlagClear()      (PIR1bits.INT0IF = 0)
00094
00126  #define EXT_INT0_InterruptDisable()      (PIE1bits.INT0IE = 0)
00127
00157  #define EXT_INT0_InterruptEnable()      (PIE1bits.INT0IE = 1)
00158
00188  #define EXT_INT0_risingEdgeSet()      (INTCON0bits.INT0EDG = 1)
00189
00219  #define EXT_INT0_fallingEdgeSet()      (INTCON0bits.INT0EDG = 0)
00248  #define EXT_INT1_InterruptFlagClear()      (PIR5bits.INT1IF = 0)
00249
00281  #define EXT_INT1_InterruptDisable()      (PIE5bits.INT1IE = 0)
00282
00312  #define EXT_INT1_InterruptEnable()      (PIE5bits.INT1IE = 1)
00313
00343  #define EXT_INT1_risingEdgeSet()      (INTCON0bits.INT1EDG = 1)
00344
00374  #define EXT_INT1_fallingEdgeSet()      (INTCON0bits.INT1EDG = 0)
00406  void EXT_INT_Initialize(void);
00407
00411
00434  void INT0_CallBack(void);
00435
00457  void INT0_SetInterruptHandler(void (* InterruptHandler)(void));
00458
00481  extern void (*INT0_InterruptHandler)(void);
00482
00505  void INT0_DefaultInterruptHandler(void);
00506
00529  void INT1_CallBack(void);
00530
00552  void INT1_SetInterruptHandler(void (* InterruptHandler)(void));
00553
00576  extern void (*INT1_InterruptHandler)(void);
00577
00600  void INT1_DefaultInterruptHandler(void);
00601
00602  // Provide C++ Compatibility
00603  #ifdef __cplusplus
00604
00605  }
00606
00607  #endif
00608  #endif

```

4.17 fvr.h

```

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00045 */
00046
00047 #ifndef FVR_H
00048 #define FVR_H
00049
00053
00054 #include <stdbool.h>
00055 #include <stdint.h>
00056
00057 #ifdef __cplusplus // Provide C++ Compatibility
00058     extern "C" {
00059
00060
00061 #endif
00062
00066
00093 void FVR_Initialize(void);
00094
00127 bool FVR_IsOutputReady(void);
00128
00129 #ifdef __cplusplus // Provide C++ Compatibility
00130
00131     }
00132
00133 #endif
00134
00135 #endif // FVR_H
00139

```

4.18 interrupt_manager.h

```

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```

00046     SOFTWARE.
00047  */
00048
00049 #ifndef INTERRUPT_MANAGER_H
00050 #define INTERRUPT_MANAGER_H
00051
00062 #define INTERRUPT_GlobalInterruptHighEnable() (INTCON0bits.GIEH = 1)
00063
00074 #define INTERRUPT_GlobalInterruptHighDisable() (INTCON0bits.GIEH = 0)
00075
00086 #define INTERRUPT_GlobalInterruptLowEnable() (INTCON0bits.GIEL = 1)
00087
00098 #define INTERRUPT_GlobalInterruptLowDisable() (INTCON0bits.GIEL = 0)
00109 void INTERRUPT_Initialize (void);
00110
00111 #endif // INTERRUPT_MANAGER_H

```

4.19 mcc.h

```

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00046     SOFTWARE.
00047  */
00048
00049 #ifndef MCC_H
00050 #define MCC_H
00051 #include <xc.h>
00052 #include "device_config.h"
00053 #include "pin_manager.h"
00054 #include <stdint.h>
00055 #include <stdbool.h>
00056 #include <conio.h>
00057 #include "interrupt_manager.h"
00058 #include "dmal.h"
00059 #include "dma2.h"
00060 #include "crc.h"
00061 #include "adcc.h"
00062 #include "fvr.h"
00063 #include "tmr5.h"
00064 #include "tmr6.h"
00065 #include "pwm8.h"
00066 #include "tmr3.h"
00067 #include "tmr2.h"
00068 #include "memory.h"
00069 #include "ext_int.h"
00070 #include "uart2.h"
00071 #include "uart1.h"
00072 #include "spi1.h"
00073 #include "clkref.h"
00074
00075 void SYSTEM_Initialize(void);
00076
00077 void OSCILLATOR_Initialize(void);
00078
00079 void PMD_Initialize(void);
00080
00081 void SystemArbiter_Initialize(void);
00082
00083 #endif /* MCC_H */

```

4.20 memory.h

```

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00046
00047 #ifndef MEMORY_H
00048 #define MEMORY_H
00049
00053
00054 #include <stdbool.h>
00055 #include <stdint.h>
00056
00057 #ifdef __cplusplus // Provide C++ Compatibility
00058
00059     extern "C" {
00060
00061 #endif
00062
00066
00067 #define WRITE_FLASH_BLOCKSIZE    128
00068 #define ERASE_FLASH_BLOCKSIZE    128
00069 #define END_FLASH                0x020000
00070
00074
00099 uint8_t FLASH_ReadByte(uint32_t flashAddr);
00100
00125 uint16_t FLASH_ReadWord(uint32_t flashAddr);
00126
00157 void FLASH_WriteByte(uint32_t flashAddr, uint8_t *flashRdBufPtr, uint8_t byte);
00158
00193 int8_t FLASH_WriteBlock(uint32_t writeAddr, uint8_t *flashWrBufPtr);
00194
00218 void FLASH_EraseBlock(uint32_t baseAddr);
00219
00223
00249 void DATAEE_WriteByte(uint16_t bAdd, uint8_t bData);
00250
00275 uint8_t DATAEE_ReadByte(uint16_t bAdd);
00276
00277 void MEMORY_Tasks(void);
00278
00279 #ifdef __cplusplus // Provide C++ Compatibility
00280
00281     }
00282
00283 #endif
00284
00285 #endif // MEMORY_H
00289

```

4.21 pin_manager.h

```

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00045 */
00046
00047 #ifndef PIN_MANAGER_H
00048 #define PIN_MANAGER_H
00049
00050
00053
00054 #include <xc.h>
00055
00056 #define INPUT    1
00057 #define OUTPUT   0
00058
00059 #define HIGH     1
00060 #define LOW      0
00061
00062 #define ANALOG   1
00063 #define DIGITAL  0
00064
00065 #define PULL_UP_ENABLED      1
00066 #define PULL_UP_DISABLED    0
00067
00068 // get/set C_BATT aliases
00069 #define C_BATT_TRIS          TRISAbits.TRISA0
00070 #define C_BATT_LAT           LATAbits.LATA0
00071 #define C_BATT_PORT          PORTAbits.RA0
00072 #define C_BATT_WPU           WPUAbits.WPUA0
00073 #define C_BATT_OD             ODCONAbits.ODCA0
00074 #define C_BATT_ANS           ANSELAbits.ANSELA0
00075 #define C_BATT_SetHigh()     do { LATAbits.LATA0 = 1; } while(0)
00076 #define C_BATT_SetLow()      do { LATAbits.LATA0 = 0; } while(0)
00077 #define C_BATT_Toggle()      do { LATAbits.LATA0 = ~LATAbits.LATA0; } while(0)
00078 #define C_BATT_GetValue()    PORTAbits.RA0
00079 #define C_BATT_SetDigitalInput() do { TRISAbits.TRISA0 = 1; } while(0)
00080 #define C_BATT_SetDigitalOutput() do { TRISAbits.TRISA0 = 0; } while(0)
00081 #define C_BATT_SetPullup()   do { WPUAbits.WPUA0 = 1; } while(0)
00082 #define C_BATT_ResetPullup() do { WPUAbits.WPUA0 = 0; } while(0)
00083 #define C_BATT_SetPushPull() do { ODCONAbits.ODCA0 = 0; } while(0)
00084 #define C_BATT_SetOpenDrain() do { ODCONAbits.ODCA0 = 1; } while(0)
00085 #define C_BATT_SetAnalogMode() do { ANSELAbits.ANSELA0 = 1; } while(0)
00086 #define C_BATT_SetDigitalMode() do { ANSELAbits.ANSELA0 = 0; } while(0)
00087
00088 // get/set C_PV aliases
00089 #define C_PV_TRIS            TRISAbits.TRISA1
00090 #define C_PV_LAT             LATAbits.LATA1
00091 #define C_PV_PORT            PORTAbits.RA1
00092 #define C_PV_WPU             WPUAbits.WPUA1
00093 #define C_PV_OD              ODCONAbits.ODCA1
00094 #define C_PV_ANS             ANSELAbits.ANSELA1
00095 #define C_PV_SetHigh()       do { LATAbits.LATA1 = 1; } while(0)
00096 #define C_PV_SetLow()        do { LATAbits.LATA1 = 0; } while(0)
00097 #define C_PV_Toggle()        do { LATAbits.LATA1 = ~LATAbits.LATA1; } while(0)
00098 #define C_PV_GetValue()      PORTAbits.RA1
00099 #define C_PV_SetDigitalInput() do { TRISAbits.TRISA1 = 1; } while(0)
00100 #define C_PV_SetDigitalOutput() do { TRISAbits.TRISA1 = 0; } while(0)
00101 #define C_PV_SetPullup()     do { WPUAbits.WPUA1 = 1; } while(0)
00102 #define C_PV_ResetPullup()   do { WPUAbits.WPUA1 = 0; } while(0)
00103 #define C_PV_SetPushPull()   do { ODCONAbits.ODCA1 = 0; } while(0)
00104 #define C_PV_SetOpenDrain()   do { ODCONAbits.ODCA1 = 1; } while(0)
00105 #define C_PV_SetAnalogMode() do { ANSELAbits.ANSELA1 = 1; } while(0)
00106 #define C_PV_SetDigitalMode() do { ANSELAbits.ANSELA1 = 0; } while(0)
00107
00108 // get/set V_CC aliases
00109 #define V_CC_TRIS            TRISAbits.TRISA2
00110 #define V_CC_LAT             LATAbits.LATA2
00111 #define V_CC_PORT            PORTAbits.RA2
00112 #define V_CC_WPU             WPUAbits.WPUA2
00113 #define V_CC_OD              ODCONAbits.ODCA2
00114 #define V_CC_ANS             ANSELAbits.ANSELA2
00115 #define V_CC_SetHigh()       do { LATAbits.LATA2 = 1; } while(0)

```



```

00116 #define V_CC_SetLow() do { LATAbits.LATA2 = 0; } while(0)
00117 #define V_CC_Toggle() do { LATAbits.LATA2 = ~LATAbits.LATA2; } while(0)
00118 #define V_CC_GetValue() PORTAbits.RA2
00119 #define V_CC_SetDigitalInput() do { TRISAbits.TRISA2 = 1; } while(0)
00120 #define V_CC_SetDigitalOutput() do { TRISAbits.TRISA2 = 0; } while(0)
00121 #define V_CC_SetPullup() do { WPUAbits.WPUA2 = 1; } while(0)
00122 #define V_CC_ResetPullup() do { WPUAbits.WPUA2 = 0; } while(0)
00123 #define V_CC_SetPushPull() do { ODCONAbits.ODCA2 = 0; } while(0)
00124 #define V_CC_SetOpenDrain() do { ODCONAbits.ODCA2 = 1; } while(0)
00125 #define V_CC_SetAnalogMode() do { ANSELAbits.ANSELA2 = 1; } while(0)
00126 #define V_CC_SetDigitalMode() do { ANSELAbits.ANSELA2 = 0; } while(0)
00127
00128 // get/set RA3 procedures
00129 #define RA3_SetHigh() do { LATAbits.LATA3 = 1; } while(0)
00130 #define RA3_SetLow() do { LATAbits.LATA3 = 0; } while(0)
00131 #define RA3_Toggle() do { LATAbits.LATA3 = ~LATAbits.LATA3; } while(0)
00132 #define RA3_GetValue() PORTAbits.RA3
00133 #define RA3_SetDigitalInput() do { TRISAbits.TRISA3 = 1; } while(0)
00134 #define RA3_SetDigitalOutput() do { TRISAbits.TRISA3 = 0; } while(0)
00135 #define RA3_SetPullup() do { WPUAbits.WPUA3 = 1; } while(0)
00136 #define RA3_ResetPullup() do { WPUAbits.WPUA3 = 0; } while(0)
00137 #define RA3_SetAnalogMode() do { ANSELAbits.ANSELA3 = 1; } while(0)
00138 #define RA3_SetDigitalMode() do { ANSELAbits.ANSELA3 = 0; } while(0)
00139
00140 // get/set V_BAT aliases
00141 #define V_BAT_TRIS TRISAbits.TRISA4
00142 #define V_BAT_LAT LATAbits.LATA4
00143 #define V_BAT_PORT PORTAbits.RA4
00144 #define V_BAT_WPU WPUAbits.WPUA4
00145 #define V_BAT_OD ODCONAbits.ODCA4
00146 #define V_BAT_ANS ANSELAbits.ANSELA4
00147 #define V_BAT_SetHigh() do { LATAbits.LATA4 = 1; } while(0)
00148 #define V_BAT_SetLow() do { LATAbits.LATA4 = 0; } while(0)
00149 #define V_BAT_Toggle() do { LATAbits.LATA4 = ~LATAbits.LATA4; } while(0)
00150 #define V_BAT_GetValue() PORTAbits.RA4
00151 #define V_BAT_SetDigitalInput() do { TRISAbits.TRISA4 = 1; } while(0)
00152 #define V_BAT_SetDigitalOutput() do { TRISAbits.TRISA4 = 0; } while(0)
00153 #define V_BAT_SetPullup() do { WPUAbits.WPUA4 = 1; } while(0)
00154 #define V_BAT_ResetPullup() do { WPUAbits.WPUA4 = 0; } while(0)
00155 #define V_BAT_SetPushPull() do { ODCONAbits.ODCA4 = 0; } while(0)
00156 #define V_BAT_SetOpenDrain() do { ODCONAbits.ODCA4 = 1; } while(0)
00157 #define V_BAT_SetAnalogMode() do { ANSELAbits.ANSELA4 = 1; } while(0)
00158 #define V_BAT_SetDigitalMode() do { ANSELAbits.ANSELA4 = 0; } while(0)
00159
00160 // get/set V_PV aliases
00161 #define V_PV_TRIS TRISAbits.TRISA5
00162 #define V_PV_LAT LATAbits.LATA5
00163 #define V_PV_PORT PORTAbits.RA5
00164 #define V_PV_WPU WPUAbits.WPUA5
00165 #define V_PV_OD ODCONAbits.ODCA5
00166 #define V_PV_ANS ANSELAbits.ANSELA5
00167 #define V_PV_SetHigh() do { LATAbits.LATA5 = 1; } while(0)
00168 #define V_PV_SetLow() do { LATAbits.LATA5 = 0; } while(0)
00169 #define V_PV_Toggle() do { LATAbits.LATA5 = ~LATAbits.LATA5; } while(0)
00170 #define V_PV_GetValue() PORTAbits.RA5
00171 #define V_PV_SetDigitalInput() do { TRISAbits.TRISA5 = 1; } while(0)
00172 #define V_PV_SetDigitalOutput() do { TRISAbits.TRISA5 = 0; } while(0)
00173 #define V_PV_SetPullup() do { WPUAbits.WPUA5 = 1; } while(0)
00174 #define V_PV_ResetPullup() do { WPUAbits.WPUA5 = 0; } while(0)
00175 #define V_PV_SetPushPull() do { ODCONAbits.ODCA5 = 0; } while(0)
00176 #define V_PV_SetOpenDrain() do { ODCONAbits.ODCA5 = 1; } while(0)
00177 #define V_PV_SetAnalogMode() do { ANSELAbits.ANSELA5 = 1; } while(0)
00178 #define V_PV_SetDigitalMode() do { ANSELAbits.ANSELA5 = 0; } while(0)
00179
00180 // get/set V_CMODE aliases
00181 #define V_CMODE_TRIS TRISAbits.TRISA6
00182 #define V_CMODE_LAT LATAbits.LATA6
00183 #define V_CMODE_PORT PORTAbits.RA6
00184 #define V_CMODE_WPU WPUAbits.WPUA6
00185 #define V_CMODE_OD ODCONAbits.ODCA6
00186 #define V_CMODE_ANS ANSELAbits.ANSELA6
00187 #define V_CMODE_SetHigh() do { LATAbits.LATA6 = 1; } while(0)
00188 #define V_CMODE_SetLow() do { LATAbits.LATA6 = 0; } while(0)
00189 #define V_CMODE_Toggle() do { LATAbits.LATA6 = ~LATAbits.LATA6; } while(0)
00190 #define V_CMODE_GetValue() PORTAbits.RA6
00191 #define V_CMODE_SetDigitalInput() do { TRISAbits.TRISA6 = 1; } while(0)
00192 #define V_CMODE_SetDigitalOutput() do { TRISAbits.TRISA6 = 0; } while(0)
00193 #define V_CMODE_SetPullup() do { WPUAbits.WPUA6 = 1; } while(0)
00194 #define V_CMODE_ResetPullup() do { WPUAbits.WPUA6 = 0; } while(0)
00195 #define V_CMODE_SetPushPull() do { ODCONAbits.ODCA6 = 0; } while(0)
00196 #define V_CMODE_SetOpenDrain() do { ODCONAbits.ODCA6 = 1; } while(0)
00197 #define V_CMODE_SetAnalogMode() do { ANSELAbits.ANSELA6 = 1; } while(0)
00198 #define V_CMODE_SetDigitalMode() do { ANSELAbits.ANSELA6 = 0; } while(0)
00199
00200 // get/set RB0 procedures
00201 #define RB0_SetHigh() do { LATBbits.LATB0 = 1; } while(0)
00202 #define RB0_SetLow() do { LATBbits.LATB0 = 0; } while(0)

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00203 #define RB0_Toggle()          do { LATBbits.LATB0 = ~LATBbits.LATB0; } while(0)
00204 #define RB0_GetValue()        PORTBbits.RB0
00205 #define RB0_SetDigitalInput() do { TRISBbits.TRISB0 = 1; } while(0)
00206 #define RB0_SetDigitalOutput() do { TRISBbits.TRISB0 = 0; } while(0)
00207 #define RB0_SetPullup()       do { WPUBbits.WPUB0 = 1; } while(0)
00208 #define RB0_ResetPullup()     do { WPUBbits.WPUB0 = 0; } while(0)
00209 #define RB0_SetAnalogMode()   do { ANSELBbits.ANSELB0 = 1; } while(0)
00210 #define RB0_SetDigitalMode()  do { ANSELBbits.ANSELB0 = 0; } while(0)
00211
00212 // get/set RB1 procedures
00213 #define RB1_SetHigh()         do { LATBbits.LATB1 = 1; } while(0)
00214 #define RB1_SetLow()         do { LATBbits.LATB1 = 0; } while(0)
00215 #define RB1_Toggle()         do { LATBbits.LATB1 = ~LATBbits.LATB1; } while(0)
00216 #define RB1_GetValue()       PORTBbits.RB1
00217 #define RB1_SetDigitalInput() do { TRISBbits.TRISB1 = 1; } while(0)
00218 #define RB1_SetDigitalOutput() do { TRISBbits.TRISB1 = 0; } while(0)
00219 #define RB1_SetPullup()      do { WPUBbits.WPUB1 = 1; } while(0)
00220 #define RB1_ResetPullup()    do { WPUBbits.WPUB1 = 0; } while(0)
00221 #define RB1_SetAnalogMode()  do { ANSELBbits.ANSELB1 = 1; } while(0)
00222 #define RB1_SetDigitalMode() do { ANSELBbits.ANSELB1 = 0; } while(0)
00223
00224 // get/set V_BBAT aliases
00225 #define V_BBAT_TRIS          TRISBbits.TRISB2
00226 #define V_BBAT_LAT           LATBbits.LATB2
00227 #define V_BBAT_PORT          PORTBbits.RB2
00228 #define V_BBAT_WPU           WPUBbits.WPUB2
00229 #define V_BBAT_OD            ODCONBbits.ODCB2
00230 #define V_BBAT_ANS           ANSELBbits.ANSELB2
00231 #define V_BBAT_SetHigh()     do { LATBbits.LATB2 = 1; } while(0)
00232 #define V_BBAT_SetLow()     do { LATBbits.LATB2 = 0; } while(0)
00233 #define V_BBAT_Toggle()     do { LATBbits.LATB2 = ~LATBbits.LATB2; } while(0)
00234 #define V_BBAT_GetValue()    PORTBbits.RB2
00235 #define V_BBAT_SetDigitalInput() do { TRISBbits.TRISB2 = 1; } while(0)
00236 #define V_BBAT_SetDigitalOutput() do { TRISBbits.TRISB2 = 0; } while(0)
00237 #define V_BBAT_SetPullup()   do { WPUBbits.WPUB2 = 1; } while(0)
00238 #define V_BBAT_ResetPullup() do { WPUBbits.WPUB2 = 0; } while(0)
00239 #define V_BBAT_SetPushPull() do { ODCONBbits.ODCB2 = 0; } while(0)
00240 #define V_BBAT_SetOpenDrain() do { ODCONBbits.ODCB2 = 1; } while(0)
00241 #define V_BBAT_SetAnalogMode() do { ANSELBbits.ANSELB2 = 1; } while(0)
00242 #define V_BBAT_SetDigitalMode() do { ANSELBbits.ANSELB2 = 0; } while(0)
00243
00244 // get/set C_MPPT aliases
00245 #define C_MPPT_TRIS          TRISBbits.TRISB3
00246 #define C_MPPT_LAT           LATBbits.LATB3
00247 #define C_MPPT_PORT          PORTBbits.RB3
00248 #define C_MPPT_WPU           WPUBbits.WPUB3
00249 #define C_MPPT_OD            ODCONBbits.ODCB3
00250 #define C_MPPT_ANS           ANSELBbits.ANSELB3
00251 #define C_MPPT_SetHigh()     do { LATBbits.LATB3 = 1; } while(0)
00252 #define C_MPPT_SetLow()     do { LATBbits.LATB3 = 0; } while(0)
00253 #define C_MPPT_Toggle()     do { LATBbits.LATB3 = ~LATBbits.LATB3; } while(0)
00254 #define C_MPPT_GetValue()    PORTBbits.RB3
00255 #define C_MPPT_SetDigitalInput() do { TRISBbits.TRISB3 = 1; } while(0)
00256 #define C_MPPT_SetDigitalOutput() do { TRISBbits.TRISB3 = 0; } while(0)
00257 #define C_MPPT_SetPullup()   do { WPUBbits.WPUB3 = 1; } while(0)
00258 #define C_MPPT_ResetPullup() do { WPUBbits.WPUB3 = 0; } while(0)
00259 #define C_MPPT_SetPushPull() do { ODCONBbits.ODCB3 = 0; } while(0)
00260 #define C_MPPT_SetOpenDrain() do { ODCONBbits.ODCB3 = 1; } while(0)
00261 #define C_MPPT_SetAnalogMode() do { ANSELBbits.ANSELB3 = 1; } while(0)
00262 #define C_MPPT_SetDigitalMode() do { ANSELBbits.ANSELB3 = 0; } while(0)
00263
00264 // get/set V_INVERTER aliases
00265 #define V_INVERTER_TRIS      TRISBbits.TRISB4
00266 #define V_INVERTER_LAT       LATBbits.LATB4
00267 #define V_INVERTER_PORT      PORTBbits.RB4
00268 #define V_INVERTER_WPU       WPUBbits.WPUB4
00269 #define V_INVERTER_OD        ODCONBbits.ODCB4
00270 #define V_INVERTER_ANS       ANSELBbits.ANSELB4
00271 #define V_INVERTER_SetHigh() do { LATBbits.LATB4 = 1; } while(0)
00272 #define V_INVERTER_SetLow() do { LATBbits.LATB4 = 0; } while(0)
00273 #define V_INVERTER_Toggle() do { LATBbits.LATB4 = ~LATBbits.LATB4; } while(0)
00274 #define V_INVERTER_GetValue() PORTBbits.RB4
00275 #define V_INVERTER_SetDigitalInput() do { TRISBbits.TRISB4 = 1; } while(0)
00276 #define V_INVERTER_SetDigitalOutput() do { TRISBbits.TRISB4 = 0; } while(0)
00277 #define V_INVERTER_SetPullup() do { WPUBbits.WPUB4 = 1; } while(0)
00278 #define V_INVERTER_ResetPullup() do { WPUBbits.WPUB4 = 0; } while(0)
00279 #define V_INVERTER_SetPushPull() do { ODCONBbits.ODCB4 = 0; } while(0)
00280 #define V_INVERTER_SetOpenDrain() do { ODCONBbits.ODCB4 = 1; } while(0)
00281 #define V_INVERTER_SetAnalogMode() do { ANSELBbits.ANSELB4 = 1; } while(0)
00282 #define V_INVERTER_SetDigitalMode() do { ANSELBbits.ANSELB4 = 0; } while(0)
00283
00284 // get/set V_LIGHT_SENSOR aliases
00285 #define V_LIGHT_SENSOR_TRIS  TRISBbits.TRISB5
00286 #define V_LIGHT_SENSOR_LAT   LATBbits.LATB5
00287 #define V_LIGHT_SENSOR_PORT  PORTBbits.RB5
00288 #define V_LIGHT_SENSOR_WPU   WPUBbits.WPUB5
00289 #define V_LIGHT_SENSOR_OD    ODCONBbits.ODCB5

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00290 #define V_LIGHT_SENSOR_ANS ANSELBbits.ANSELB5
00291 #define V_LIGHT_SENSOR_SetHigh() do { LATBbits.LATB5 = 1; } while(0)
00292 #define V_LIGHT_SENSOR_SetLow() do { LATBbits.LATB5 = 0; } while(0)
00293 #define V_LIGHT_SENSOR_Toggle() do { LATBbits.LATB5 = ~LATBbits.LATB5; } while(0)
00294 #define V_LIGHT_SENSOR_GetValue() PORTBbits.RB5
00295 #define V_LIGHT_SENSOR_SetDigitalInput() do { TRISBbits.TRISB5 = 1; } while(0)
00296 #define V_LIGHT_SENSOR_SetDigitalOutput() do { TRISBbits.TRISB5 = 0; } while(0)
00297 #define V_LIGHT_SENSOR_SetPullup() do { WPUBbits.WPUB5 = 1; } while(0)
00298 #define V_LIGHT_SENSOR_ResetPullup() do { WPUBbits.WPUB5 = 0; } while(0)
00299 #define V_LIGHT_SENSOR_SetPushPull() do { ODCONBbits.ODCB5 = 0; } while(0)
00300 #define V_LIGHT_SENSOR_SetOpenDrain() do { ODCONBbits.ODCB5 = 1; } while(0)
00301 #define V_LIGHT_SENSOR_SetAnalogMode() do { ANSELBbits.ANSELB5 = 1; } while(0)
00302 #define V_LIGHT_SENSOR_SetDigitalMode() do { ANSELBbits.ANSELB5 = 0; } while(0)
00303
00304 // get/set RB6 procedures
00305 #define RB6_SetHigh() do { LATBbits.LATB6 = 1; } while(0)
00306 #define RB6_SetLow() do { LATBbits.LATB6 = 0; } while(0)
00307 #define RB6_Toggle() do { LATBbits.LATB6 = ~LATBbits.LATB6; } while(0)
00308 #define RB6_GetValue() PORTBbits.RB6
00309 #define RB6_SetDigitalInput() do { TRISBbits.TRISB6 = 1; } while(0)
00310 #define RB6_SetDigitalOutput() do { TRISBbits.TRISB6 = 0; } while(0)
00311 #define RB6_SetPullup() do { WPUBbits.WPUB6 = 1; } while(0)
00312 #define RB6_ResetPullup() do { WPUBbits.WPUB6 = 0; } while(0)
00313 #define RB6_SetAnalogMode() do { ANSELBbits.ANSELB6 = 1; } while(0)
00314 #define RB6_SetDigitalMode() do { ANSELBbits.ANSELB6 = 0; } while(0)
00315
00316 // get/set RB7 procedures
00317 #define RB7_SetHigh() do { LATBbits.LATB7 = 1; } while(0)
00318 #define RB7_SetLow() do { LATBbits.LATB7 = 0; } while(0)
00319 #define RB7_Toggle() do { LATBbits.LATB7 = ~LATBbits.LATB7; } while(0)
00320 #define RB7_GetValue() PORTBbits.RB7
00321 #define RB7_SetDigitalInput() do { TRISBbits.TRISB7 = 1; } while(0)
00322 #define RB7_SetDigitalOutput() do { TRISBbits.TRISB7 = 0; } while(0)
00323 #define RB7_SetPullup() do { WPUBbits.WPUB7 = 1; } while(0)
00324 #define RB7_ResetPullup() do { WPUBbits.WPUB7 = 0; } while(0)
00325 #define RB7_SetAnalogMode() do { ANSELBbits.ANSELB7 = 1; } while(0)
00326 #define RB7_SetDigitalMode() do { ANSELBbits.ANSELB7 = 0; } while(0)
00327
00328 // get/set CS_SDCARD aliases
00329 #define CS_SDCARD_TRIS TRISCbits.TRISC0
00330 #define CS_SDCARD_LAT LATCbits.LATC0
00331 #define CS_SDCARD_PORT PORTCbits.RC0
00332 #define CS_SDCARD_WPU WPUCbits.WPUC0
00333 #define CS_SDCARD_OD ODCONCbits.ODCC0
00334 #define CS_SDCARD_ANS ANSELCbits.ANSELC0
00335 #define CS_SDCARD_SetHigh() do { LATCbits.LATC0 = 1; } while(0)
00336 #define CS_SDCARD_SetLow() do { LATCbits.LATC0 = 0; } while(0)
00337 #define CS_SDCARD_Toggle() do { LATCbits.LATC0 = ~LATCbits.LATC0; } while(0)
00338 #define CS_SDCARD_GetValue() PORTCbits.RC0
00339 #define CS_SDCARD_SetDigitalInput() do { TRISCbits.TRISC0 = 1; } while(0)
00340 #define CS_SDCARD_SetDigitalOutput() do { TRISCbits.TRISC0 = 0; } while(0)
00341 #define CS_SDCARD_SetPullup() do { WPUCbits.WPUC0 = 1; } while(0)
00342 #define CS_SDCARD_ResetPullup() do { WPUCbits.WPUC0 = 0; } while(0)
00343 #define CS_SDCARD_SetPushPull() do { ODCONCbits.ODCC0 = 0; } while(0)
00344 #define CS_SDCARD_SetOpenDrain() do { ODCONCbits.ODCC0 = 1; } while(0)
00345 #define CS_SDCARD_SetAnalogMode() do { ANSELCbits.ANSELC0 = 1; } while(0)
00346 #define CS_SDCARD_SetDigitalMode() do { ANSELCbits.ANSELC0 = 0; } while(0)
00347
00348 // get/set RS aliases
00349 #define RS_TRIS TRISCbits.TRISC1
00350 #define RS_LAT LATCbits.LATC1
00351 #define RS_PORT PORTCbits.RC1
00352 #define RS_WPU WPUCbits.WPUC1
00353 #define RS_OD ODCONCbits.ODCC1
00354 #define RS_ANS ANSELCbits.ANSELC1
00355 #define RS_SetHigh() do { LATCbits.LATC1 = 1; } while(0)
00356 #define RS_SetLow() do { LATCbits.LATC1 = 0; } while(0)
00357 #define RS_Toggle() do { LATCbits.LATC1 = ~LATCbits.LATC1; } while(0)
00358 #define RS_GetValue() PORTCbits.RC1
00359 #define RS_SetDigitalInput() do { TRISCbits.TRISC1 = 1; } while(0)
00360 #define RS_SetDigitalOutput() do { TRISCbits.TRISC1 = 0; } while(0)
00361 #define RS_SetPullup() do { WPUCbits.WPUC1 = 1; } while(0)
00362 #define RS_ResetPullup() do { WPUCbits.WPUC1 = 0; } while(0)
00363 #define RS_SetPushPull() do { ODCONCbits.ODCC1 = 0; } while(0)
00364 #define RS_SetOpenDrain() do { ODCONCbits.ODCC1 = 1; } while(0)
00365 #define RS_SetAnalogMode() do { ANSELCbits.ANSELC1 = 1; } while(0)
00366 #define RS_SetDigitalMode() do { ANSELCbits.ANSELC1 = 0; } while(0)
00367
00368 // get/set CSB aliases
00369 #define CSB_TRIS TRISCbits.TRISC2
00370 #define CSB_LAT LATCbits.LATC2
00371 #define CSB_PORT PORTCbits.RC2
00372 #define CSB_WPU WPUCbits.WPUC2
00373 #define CSB_OD ODCONCbits.ODCC2
00374 #define CSB_ANS ANSELCbits.ANSELC2
00375 #define CSB_SetHigh() do { LATCbits.LATC2 = 1; } while(0)
00376 #define CSB_SetLow() do { LATCbits.LATC2 = 0; } while(0)

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00377 #define CSB_Toggle() do { LATCbits.LATC2 = ~LATCbits.LATC2; } while(0)
00378 #define CSB_GetValue() PORTCbits.RC2
00379 #define CSB_SetDigitalInput() do { TRISCbits.TRISC2 = 1; } while(0)
00380 #define CSB_SetDigitalOutput() do { TRISCbits.TRISC2 = 0; } while(0)
00381 #define CSB_SetPullup() do { WPUCbits.WPUC2 = 1; } while(0)
00382 #define CSB_ResetPullup() do { WPUCbits.WPUC2 = 0; } while(0)
00383 #define CSB_SetPushPull() do { ODCONCbits.ODCC2 = 0; } while(0)
00384 #define CSB_SetOpenDrain() do { ODCONCbits.ODCC2 = 1; } while(0)
00385 #define CSB_SetAnalogMode() do { ANSELcbits.ANSELc2 = 1; } while(0)
00386 #define CSB_SetDigitalMode() do { ANSELcbits.ANSELc2 = 0; } while(0)
00387
00388 // get/set RC3 procedures
00389 #define RC3_SetHigh() do { LATCbits.LATC3 = 1; } while(0)
00390 #define RC3_SetLow() do { LATCbits.LATC3 = 0; } while(0)
00391 #define RC3_Toggle() do { LATCbits.LATC3 = ~LATCbits.LATC3; } while(0)
00392 #define RC3_GetValue() PORTCbits.RC3
00393 #define RC3_SetDigitalInput() do { TRISCbits.TRISC3 = 1; } while(0)
00394 #define RC3_SetDigitalOutput() do { TRISCbits.TRISC3 = 0; } while(0)
00395 #define RC3_SetPullup() do { WPUCbits.WPUC3 = 1; } while(0)
00396 #define RC3_ResetPullup() do { WPUCbits.WPUC3 = 0; } while(0)
00397 #define RC3_SetAnalogMode() do { ANSELcbits.ANSELc3 = 1; } while(0)
00398 #define RC3_SetDigitalMode() do { ANSELcbits.ANSELc3 = 0; } while(0)
00399
00400 // get/set RC4 procedures
00401 #define RC4_SetHigh() do { LATCbits.LATC4 = 1; } while(0)
00402 #define RC4_SetLow() do { LATCbits.LATC4 = 0; } while(0)
00403 #define RC4_Toggle() do { LATCbits.LATC4 = ~LATCbits.LATC4; } while(0)
00404 #define RC4_GetValue() PORTCbits.RC4
00405 #define RC4_SetDigitalInput() do { TRISCbits.TRISC4 = 1; } while(0)
00406 #define RC4_SetDigitalOutput() do { TRISCbits.TRISC4 = 0; } while(0)
00407 #define RC4_SetPullup() do { WPUCbits.WPUC4 = 1; } while(0)
00408 #define RC4_ResetPullup() do { WPUCbits.WPUC4 = 0; } while(0)
00409 #define RC4_SetAnalogMode() do { ANSELcbits.ANSELc4 = 1; } while(0)
00410 #define RC4_SetDigitalMode() do { ANSELcbits.ANSELc4 = 0; } while(0)
00411
00412 // get/set RC5 procedures
00413 #define RC5_SetHigh() do { LATCbits.LATC5 = 1; } while(0)
00414 #define RC5_SetLow() do { LATCbits.LATC5 = 0; } while(0)
00415 #define RC5_Toggle() do { LATCbits.LATC5 = ~LATCbits.LATC5; } while(0)
00416 #define RC5_GetValue() PORTCbits.RC5
00417 #define RC5_SetDigitalInput() do { TRISCbits.TRISC5 = 1; } while(0)
00418 #define RC5_SetDigitalOutput() do { TRISCbits.TRISC5 = 0; } while(0)
00419 #define RC5_SetPullup() do { WPUCbits.WPUC5 = 1; } while(0)
00420 #define RC5_ResetPullup() do { WPUCbits.WPUC5 = 0; } while(0)
00421 #define RC5_SetAnalogMode() do { ANSELcbits.ANSELc5 = 1; } while(0)
00422 #define RC5_SetDigitalMode() do { ANSELcbits.ANSELc5 = 0; } while(0)
00423
00424 // get/set RC6 procedures
00425 #define RC6_SetHigh() do { LATCbits.LATC6 = 1; } while(0)
00426 #define RC6_SetLow() do { LATCbits.LATC6 = 0; } while(0)
00427 #define RC6_Toggle() do { LATCbits.LATC6 = ~LATCbits.LATC6; } while(0)
00428 #define RC6_GetValue() PORTCbits.RC6
00429 #define RC6_SetDigitalInput() do { TRISCbits.TRISC6 = 1; } while(0)
00430 #define RC6_SetDigitalOutput() do { TRISCbits.TRISC6 = 0; } while(0)
00431 #define RC6_SetPullup() do { WPUCbits.WPUC6 = 1; } while(0)
00432 #define RC6_ResetPullup() do { WPUCbits.WPUC6 = 0; } while(0)
00433 #define RC6_SetAnalogMode() do { ANSELcbits.ANSELc6 = 1; } while(0)
00434 #define RC6_SetDigitalMode() do { ANSELcbits.ANSELc6 = 0; } while(0)
00435
00436 // get/set RC7 procedures
00437 #define RC7_SetHigh() do { LATCbits.LATC7 = 1; } while(0)
00438 #define RC7_SetLow() do { LATCbits.LATC7 = 0; } while(0)
00439 #define RC7_Toggle() do { LATCbits.LATC7 = ~LATCbits.LATC7; } while(0)
00440 #define RC7_GetValue() PORTCbits.RC7
00441 #define RC7_SetDigitalInput() do { TRISCbits.TRISC7 = 1; } while(0)
00442 #define RC7_SetDigitalOutput() do { TRISCbits.TRISC7 = 0; } while(0)
00443 #define RC7_SetPullup() do { WPUCbits.WPUC7 = 1; } while(0)
00444 #define RC7_ResetPullup() do { WPUCbits.WPUC7 = 0; } while(0)
00445 #define RC7_SetAnalogMode() do { ANSELcbits.ANSELc7 = 1; } while(0)
00446 #define RC7_SetDigitalMode() do { ANSELcbits.ANSELc7 = 0; } while(0)
00447
00448 // get/set RELAY0 aliases
00449 #define RELAY0_TRIS TRISDbits.TRISD0
00450 #define RELAY0_LAT LATDbits.LATD0
00451 #define RELAY0_PORT PORTDbits.RD0
00452 #define RELAY0_WPU WPUDbits.WPUD0
00453 #define RELAY0_OD ODCONDbits.ODCD0
00454 #define RELAY0_ANS ANSELDbits.ANSELD0
00455 #define RELAY0_SetHigh() do { LATDbits.LATD0 = 1; } while(0)
00456 #define RELAY0_SetLow() do { LATDbits.LATD0 = 0; } while(0)
00457 #define RELAY0_Toggle() do { LATDbits.LATD0 = ~LATDbits.LATD0; } while(0)
00458 #define RELAY0_GetValue() PORTDbits.RD0
00459 #define RELAY0_SetDigitalInput() do { TRISDbits.TRISD0 = 1; } while(0)
00460 #define RELAY0_SetDigitalOutput() do { TRISDbits.TRISD0 = 0; } while(0)
00461 #define RELAY0_SetPullup() do { WPUDbits.WPUD0 = 1; } while(0)
00462 #define RELAY0_ResetPullup() do { WPUDbits.WPUD0 = 0; } while(0)
00463 #define RELAY0_SetPushPull() do { ODCONDbits.ODCD0 = 0; } while(0)

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00464 #define RELAY0_SetOpenDrain()      do { ODCONDbits.ODCD0 = 1; } while(0)
00465 #define RELAY0_SetAnalogMode()     do { ANSELDbits.ANSELD0 = 1; } while(0)
00466 #define RELAY0_SetDigitalMode()    do { ANSELDbits.ANSELD0 = 0; } while(0)
00467
00468 // get/set RD1 procedures
00469 #define RD1_SetHigh()              do { LATDbits.LATD1 = 1; } while(0)
00470 #define RD1_SetLow()               do { LATDbits.LATD1 = 0; } while(0)
00471 #define RD1_Toggle()               do { LATDbits.LATD1 = ~LATDbits.LATD1; } while(0)
00472 #define RD1_GetValue()             PORTDbits.RD1
00473 #define RD1_SetDigitalInput()      do { TRISDbits.TRISD1 = 1; } while(0)
00474 #define RD1_SetDigitalOutput()     do { TRISDbits.TRISD1 = 0; } while(0)
00475 #define RD1_SetPullup()            do { WPUDbits.WPUD1 = 1; } while(0)
00476 #define RD1_ResetPullup()          do { WPUDbits.WPUD1 = 0; } while(0)
00477 #define RD1_SetAnalogMode()        do { ANSELDbits.ANSELD1 = 1; } while(0)
00478 #define RD1_SetDigitalMode()       do { ANSELDbits.ANSELD1 = 0; } while(0)
00479
00480 // get/set SW0 aliases
00481 #define SW0_TRIS                   TRISDbits.TRISD2
00482 #define SW0_LAT                     LATDbits.LATD2
00483 #define SW0_PORT                   PORTDbits.RD2
00484 #define SW0_WPU                    WPUDbits.WPUD2
00485 #define SW0_OD                     ODCONDbits.ODCD2
00486 #define SW0_ANS                     ANSELDbits.ANSELD2
00487 #define SW0_SetHigh()              do { LATDbits.LATD2 = 1; } while(0)
00488 #define SW0_SetLow()               do { LATDbits.LATD2 = 0; } while(0)
00489 #define SW0_Toggle()               do { LATDbits.LATD2 = ~LATDbits.LATD2; } while(0)
00490 #define SW0_GetValue()             PORTDbits.RD2
00491 #define SW0_SetDigitalInput()      do { TRISDbits.TRISD2 = 1; } while(0)
00492 #define SW0_SetDigitalOutput()     do { TRISDbits.TRISD2 = 0; } while(0)
00493 #define SW0_SetPullup()            do { WPUDbits.WPUD2 = 1; } while(0)
00494 #define SW0_ResetPullup()          do { WPUDbits.WPUD2 = 0; } while(0)
00495 #define SW0_SetPushPull()          do { ODCONDbits.ODCD2 = 0; } while(0)
00496 #define SW0_SetOpenDrain()         do { ODCONDbits.ODCD2 = 1; } while(0)
00497 #define SW0_SetAnalogMode()        do { ANSELDbits.ANSELD2 = 1; } while(0)
00498 #define SW0_SetDigitalMode()       do { ANSELDbits.ANSELD2 = 0; } while(0)
00499
00500 // get/set CALIB aliases
00501 #define CALIB_TRIS                 TRISDbits.TRISD3
00502 #define CALIB_LAT                   LATDbits.LATD3
00503 #define CALIB_PORT                 PORTDbits.RD3
00504 #define CALIB_WPU                  WPUDbits.WPUD3
00505 #define CALIB_OD                   ODCONDbits.ODCD3
00506 #define CALIB_ANS                   ANSELDbits.ANSELD3
00507 #define CALIB_SetHigh()            do { LATDbits.LATD3 = 1; } while(0)
00508 #define CALIB_SetLow()             do { LATDbits.LATD3 = 0; } while(0)
00509 #define CALIB_Toggle()             do { LATDbits.LATD3 = ~LATDbits.LATD3; } while(0)
00510 #define CALIB_GetValue()           PORTDbits.RD3
00511 #define CALIB_SetDigitalInput()     do { TRISDbits.TRISD3 = 1; } while(0)
00512 #define CALIB_SetDigitalOutput()    do { TRISDbits.TRISD3 = 0; } while(0)
00513 #define CALIB_SetPullup()          do { WPUDbits.WPUD3 = 1; } while(0)
00514 #define CALIB_ResetPullup()         do { WPUDbits.WPUD3 = 0; } while(0)
00515 #define CALIB_SetPushPull()         do { ODCONDbits.ODCD3 = 0; } while(0)
00516 #define CALIB_SetOpenDrain()        do { ODCONDbits.ODCD3 = 1; } while(0)
00517 #define CALIB_SetAnalogMode()       do { ANSELDbits.ANSELD3 = 1; } while(0)
00518 #define CALIB_SetDigitalMode()      do { ANSELDbits.ANSELD3 = 0; } while(0)
00519
00520 // get/set RELAYL1 aliases
00521 #define RELAYL1_TRIS                TRISDbits.TRISD4
00522 #define RELAYL1_LAT                 LATDbits.LATD4
00523 #define RELAYL1_PORT               PORTDbits.RD4
00524 #define RELAYL1_WPU                WPUDbits.WPUD4
00525 #define RELAYL1_OD                 ODCONDbits.ODCD4
00526 #define RELAYL1_ANS                 ANSELDbits.ANSELD4
00527 #define RELAYL1_SetHigh()          do { LATDbits.LATD4 = 1; } while(0)
00528 #define RELAYL1_SetLow()           do { LATDbits.LATD4 = 0; } while(0)
00529 #define RELAYL1_Toggle()           do { LATDbits.LATD4 = ~LATDbits.LATD4; } while(0)
00530 #define RELAYL1_GetValue()         PORTDbits.RD4
00531 #define RELAYL1_SetDigitalInput()   do { TRISDbits.TRISD4 = 1; } while(0)
00532 #define RELAYL1_SetDigitalOutput()  do { TRISDbits.TRISD4 = 0; } while(0)
00533 #define RELAYL1_SetPullup()        do { WPUDbits.WPUD4 = 1; } while(0)
00534 #define RELAYL1_ResetPullup()      do { WPUDbits.WPUD4 = 0; } while(0)
00535 #define RELAYL1_SetPushPull()       do { ODCONDbits.ODCD4 = 0; } while(0)
00536 #define RELAYL1_SetOpenDrain()      do { ODCONDbits.ODCD4 = 1; } while(0)
00537 #define RELAYL1_SetAnalogMode()     do { ANSELDbits.ANSELD4 = 1; } while(0)
00538 #define RELAYL1_SetDigitalMode()    do { ANSELDbits.ANSELD4 = 0; } while(0)
00539
00540 // get/set RELAYL2 aliases
00541 #define RELAYL2_TRIS                TRISDbits.TRISD5
00542 #define RELAYL2_LAT                 LATDbits.LATD5
00543 #define RELAYL2_PORT               PORTDbits.RD5
00544 #define RELAYL2_WPU                WPUDbits.WPUD5
00545 #define RELAYL2_OD                 ODCONDbits.ODCD5
00546 #define RELAYL2_ANS                 ANSELDbits.ANSELD5
00547 #define RELAYL2_SetHigh()          do { LATDbits.LATD5 = 1; } while(0)
00548 #define RELAYL2_SetLow()           do { LATDbits.LATD5 = 0; } while(0)
00549 #define RELAYL2_Toggle()           do { LATDbits.LATD5 = ~LATDbits.LATD5; } while(0)
00550 #define RELAYL2_GetValue()         PORTDbits.RD5

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00551 #define RELAYL2_SetDigitalInput() do { TRISDbits.TRISD5 = 1; } while(0)
00552 #define RELAYL2_SetDigitalOutput() do { TRISDbits.TRISD5 = 0; } while(0)
00553 #define RELAYL2_SetPullup() do { WPUDbits.WPUD5 = 1; } while(0)
00554 #define RELAYL2_ResetPullup() do { WPUDbits.WPUD5 = 0; } while(0)
00555 #define RELAYL2_SetPushPull() do { ODCONDbits.ODCD5 = 0; } while(0)
00556 #define RELAYL2_SetOpenDrain() do { ODCONDbits.ODCD5 = 1; } while(0)
00557 #define RELAYL2_SetAnalogMode() do { ANSELDbits.ANSELD5 = 1; } while(0)
00558 #define RELAYL2_SetDigitalMode() do { ANSELDbits.ANSELD5 = 0; } while(0)
00559
00560 // get/set AC_CHARGER aliases
00561 #define AC_CHARGER_TRIS TRISDbits.TRISD6
00562 #define AC_CHARGER_LAT LATDbits.LATD6
00563 #define AC_CHARGER_PORT PORTDbits.RD6
00564 #define AC_CHARGER_WPU WPUDbits.WPUD6
00565 #define AC_CHARGER_OD ODCONDbits.ODCD6
00566 #define AC_CHARGER_ANS ANSELDbits.ANSELD6
00567 #define AC_CHARGER_SetHigh() do { LATDbits.LATD6 = 1; } while(0)
00568 #define AC_CHARGER_SetLow() do { LATDbits.LATD6 = 0; } while(0)
00569 #define AC_CHARGER_Toggle() do { LATDbits.LATD6 = ~LATDbits.LATD6; } while(0)
00570 #define AC_CHARGER_GetValue() PORTDbits.RD6
00571 #define AC_CHARGER_SetDigitalInput() do { TRISDbits.TRISD6 = 1; } while(0)
00572 #define AC_CHARGER_SetDigitalOutput() do { TRISDbits.TRISD6 = 0; } while(0)
00573 #define AC_CHARGER_SetPullup() do { WPUDbits.WPUD6 = 1; } while(0)
00574 #define AC_CHARGER_ResetPullup() do { WPUDbits.WPUD6 = 0; } while(0)
00575 #define AC_CHARGER_SetPushPull() do { ODCONDbits.ODCD6 = 0; } while(0)
00576 #define AC_CHARGER_SetOpenDrain() do { ODCONDbits.ODCD6 = 1; } while(0)
00577 #define AC_CHARGER_SetAnalogMode() do { ANSELDbits.ANSELD6 = 1; } while(0)
00578 #define AC_CHARGER_SetDigitalMode() do { ANSELDbits.ANSELD6 = 0; } while(0)
00579
00580 // get/set OUTP_3 aliases
00581 #define OUTP_3_TRIS TRISDbits.TRISD7
00582 #define OUTP_3_LAT LATDbits.LATD7
00583 #define OUTP_3_PORT PORTDbits.RD7
00584 #define OUTP_3_WPU WPUDbits.WPUD7
00585 #define OUTP_3_OD ODCONDbits.ODCD7
00586 #define OUTP_3_ANS ANSELDbits.ANSELD7
00587 #define OUTP_3_SetHigh() do { LATDbits.LATD7 = 1; } while(0)
00588 #define OUTP_3_SetLow() do { LATDbits.LATD7 = 0; } while(0)
00589 #define OUTP_3_Toggle() do { LATDbits.LATD7 = ~LATDbits.LATD7; } while(0)
00590 #define OUTP_3_GetValue() PORTDbits.RD7
00591 #define OUTP_3_SetDigitalInput() do { TRISDbits.TRISD7 = 1; } while(0)
00592 #define OUTP_3_SetDigitalOutput() do { TRISDbits.TRISD7 = 0; } while(0)
00593 #define OUTP_3_SetPullup() do { WPUDbits.WPUD7 = 1; } while(0)
00594 #define OUTP_3_ResetPullup() do { WPUDbits.WPUD7 = 0; } while(0)
00595 #define OUTP_3_SetPushPull() do { ODCONDbits.ODCD7 = 0; } while(0)
00596 #define OUTP_3_SetOpenDrain() do { ODCONDbits.ODCD7 = 1; } while(0)
00597 #define OUTP_3_SetAnalogMode() do { ANSELDbits.ANSELD7 = 1; } while(0)
00598 #define OUTP_3_SetDigitalMode() do { ANSELDbits.ANSELD7 = 0; } while(0)
00599
00600 // get/set LED0 aliases
00601 #define LED0_TRIS TRISEbits.TRISE0
00602 #define LED0_LAT LATEbits.LATE0
00603 #define LED0_PORT PORTEbits.RE0
00604 #define LED0_WPU WPUEbits.WPUE0
00605 #define LED0_OD ODCONEbits.ODCE0
00606 #define LED0_ANS ANSELEbits.ANSELE0
00607 #define LED0_SetHigh() do { LATEbits.LATE0 = 1; } while(0)
00608 #define LED0_SetLow() do { LATEbits.LATE0 = 0; } while(0)
00609 #define LED0_Toggle() do { LATEbits.LATE0 = ~LATEbits.LATE0; } while(0)
00610 #define LED0_GetValue() PORTEbits.RE0
00611 #define LED0_SetDigitalInput() do { TRISEbits.TRISE0 = 1; } while(0)
00612 #define LED0_SetDigitalOutput() do { TRISEbits.TRISE0 = 0; } while(0)
00613 #define LED0_SetPullup() do { WPUEbits.WPUE0 = 1; } while(0)
00614 #define LED0_ResetPullup() do { WPUEbits.WPUE0 = 0; } while(0)
00615 #define LED0_SetPushPull() do { ODCONEbits.ODCE0 = 0; } while(0)
00616 #define LED0_SetOpenDrain() do { ODCONEbits.ODCE0 = 1; } while(0)
00617 #define LED0_SetAnalogMode() do { ANSELEbits.ANSELE0 = 1; } while(0)
00618 #define LED0_SetDigitalMode() do { ANSELEbits.ANSELE0 = 0; } while(0)
00619
00620 // get/set DEBUG1 aliases
00621 #define DEBUG1_TRIS TRISEbits.TRISE1
00622 #define DEBUG1_LAT LATEbits.LATE1
00623 #define DEBUG1_PORT PORTEbits.RE1
00624 #define DEBUG1_WPU WPUEbits.WPUE1
00625 #define DEBUG1_OD ODCONEbits.ODCE1
00626 #define DEBUG1_ANS ANSELEbits.ANSELE1
00627 #define DEBUG1_SetHigh() do { LATEbits.LATE1 = 1; } while(0)
00628 #define DEBUG1_SetLow() do { LATEbits.LATE1 = 0; } while(0)
00629 #define DEBUG1_Toggle() do { LATEbits.LATE1 = ~LATEbits.LATE1; } while(0)
00630 #define DEBUG1_GetValue() PORTEbits.RE1
00631 #define DEBUG1_SetDigitalInput() do { TRISEbits.TRISE1 = 1; } while(0)
00632 #define DEBUG1_SetDigitalOutput() do { TRISEbits.TRISE1 = 0; } while(0)
00633 #define DEBUG1_SetPullup() do { WPUEbits.WPUE1 = 1; } while(0)
00634 #define DEBUG1_ResetPullup() do { WPUEbits.WPUE1 = 0; } while(0)
00635 #define DEBUG1_SetPushPull() do { ODCONEbits.ODCE1 = 0; } while(0)
00636 #define DEBUG1_SetOpenDrain() do { ODCONEbits.ODCE1 = 1; } while(0)
00637 #define DEBUG1_SetAnalogMode() do { ANSELEbits.ANSELE1 = 1; } while(0)

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00638 #define DEBUG1_SetDigitalMode() do { ANSELEbits.ANSELE1 = 0; } while(0)
00639
00640 // get/set DEBUG2 aliases
00641 #define DEBUG2_TRIS TRISEbits.TRISE2
00642 #define DEBUG2_LAT LATEbits.LATE2
00643 #define DEBUG2_PORT PORTEbits.RE2
00644 #define DEBUG2_WPU WPUEbits.WPUE2
00645 #define DEBUG2_OD ODCONEbits.ODCE2
00646 #define DEBUG2_ANS ANSELEbits.ANSELE2
00647 #define DEBUG2_SetHigh() do { LATEbits.LATE2 = 1; } while(0)
00648 #define DEBUG2_SetLow() do { LATEbits.LATE2 = 0; } while(0)
00649 #define DEBUG2_Toggle() do { LATEbits.LATE2 = ~LATEbits.LATE2; } while(0)
00650 #define DEBUG2_GetValue() PORTEbits.RE2
00651 #define DEBUG2_SetDigitalInput() do { TRISEbits.TRISE2 = 1; } while(0)
00652 #define DEBUG2_SetDigitalOutput() do { TRISEbits.TRISE2 = 0; } while(0)
00653 #define DEBUG2_SetPullup() do { WPUEbits.WPUE2 = 1; } while(0)
00654 #define DEBUG2_ResetPullup() do { WPUEbits.WPUE2 = 0; } while(0)
00655 #define DEBUG2_SetPushPull() do { ODCONEbits.ODCE2 = 0; } while(0)
00656 #define DEBUG2_SetOpenDrain() do { ODCONEbits.ODCE2 = 1; } while(0)
00657 #define DEBUG2_SetAnalogMode() do { ANSELEbits.ANSELE2 = 1; } while(0)
00658 #define DEBUG2_SetDigitalMode() do { ANSELEbits.ANSELE2 = 0; } while(0)
00659
00660 // get/set DAC_CS0 aliases
00661 #define DAC_CS0_TRIS TRISFbits.TRISF0
00662 #define DAC_CS0_LAT LATFbits.LATF0
00663 #define DAC_CS0_PORT PORTFbits.RF0
00664 #define DAC_CS0_WPU WPUFbits.WPUF0
00665 #define DAC_CS0_OD ODCONFbits.ODCF0
00666 #define DAC_CS0_ANS ANSELFbits.ANSELF0
00667 #define DAC_CS0_SetHigh() do { LATFbits.LATF0 = 1; } while(0)
00668 #define DAC_CS0_SetLow() do { LATFbits.LATF0 = 0; } while(0)
00669 #define DAC_CS0_Toggle() do { LATFbits.LATF0 = ~LATFbits.LATF0; } while(0)
00670 #define DAC_CS0_GetValue() PORTFbits.RF0
00671 #define DAC_CS0_SetDigitalInput() do { TRISFbits.TRISF0 = 1; } while(0)
00672 #define DAC_CS0_SetDigitalOutput() do { TRISFbits.TRISF0 = 0; } while(0)
00673 #define DAC_CS0_SetPullup() do { WPUFbits.WPUF0 = 1; } while(0)
00674 #define DAC_CS0_ResetPullup() do { WPUFbits.WPUF0 = 0; } while(0)
00675 #define DAC_CS0_SetPushPull() do { ODCONFbits.ODCF0 = 0; } while(0)
00676 #define DAC_CS0_SetOpenDrain() do { ODCONFbits.ODCF0 = 1; } while(0)
00677 #define DAC_CS0_SetAnalogMode() do { ANSELFbits.ANSELF0 = 1; } while(0)
00678 #define DAC_CS0_SetDigitalMode() do { ANSELFbits.ANSELF0 = 0; } while(0)
00679
00680 // get/set IO_RF1 aliases
00681 #define IO_RF1_TRIS TRISFbits.TRISF1
00682 #define IO_RF1_LAT LATFbits.LATF1
00683 #define IO_RF1_PORT PORTFbits.RF1
00684 #define IO_RF1_WPU WPUFbits.WPUF1
00685 #define IO_RF1_OD ODCONFbits.ODCF1
00686 #define IO_RF1_ANS ANSELFbits.ANSELF1
00687 #define IO_RF1_SetHigh() do { LATFbits.LATF1 = 1; } while(0)
00688 #define IO_RF1_SetLow() do { LATFbits.LATF1 = 0; } while(0)
00689 #define IO_RF1_Toggle() do { LATFbits.LATF1 = ~LATFbits.LATF1; } while(0)
00690 #define IO_RF1_GetValue() PORTFbits.RF1
00691 #define IO_RF1_SetDigitalInput() do { TRISFbits.TRISF1 = 1; } while(0)
00692 #define IO_RF1_SetDigitalOutput() do { TRISFbits.TRISF1 = 0; } while(0)
00693 #define IO_RF1_SetPullup() do { WPUFbits.WPUF1 = 1; } while(0)
00694 #define IO_RF1_ResetPullup() do { WPUFbits.WPUF1 = 0; } while(0)
00695 #define IO_RF1_SetPushPull() do { ODCONFbits.ODCF1 = 0; } while(0)
00696 #define IO_RF1_SetOpenDrain() do { ODCONFbits.ODCF1 = 1; } while(0)
00697 #define IO_RF1_SetAnalogMode() do { ANSELFbits.ANSELF1 = 1; } while(0)
00698 #define IO_RF1_SetDigitalMode() do { ANSELFbits.ANSELF1 = 0; } while(0)
00699
00700 // get/set SELECT_B aliases
00701 #define SELECT_B_TRIS TRISFbits.TRISF2
00702 #define SELECT_B_LAT LATFbits.LATF2
00703 #define SELECT_B_PORT PORTFbits.RF2
00704 #define SELECT_B_WPU WPUFbits.WPUF2
00705 #define SELECT_B_OD ODCONFbits.ODCF2
00706 #define SELECT_B_ANS ANSELFbits.ANSELF2
00707 #define SELECT_B_SetHigh() do { LATFbits.LATF2 = 1; } while(0)
00708 #define SELECT_B_SetLow() do { LATFbits.LATF2 = 0; } while(0)
00709 #define SELECT_B_Toggle() do { LATFbits.LATF2 = ~LATFbits.LATF2; } while(0)
00710 #define SELECT_B_GetValue() PORTFbits.RF2
00711 #define SELECT_B_SetDigitalInput() do { TRISFbits.TRISF2 = 1; } while(0)
00712 #define SELECT_B_SetDigitalOutput() do { TRISFbits.TRISF2 = 0; } while(0)
00713 #define SELECT_B_SetPullup() do { WPUFbits.WPUF2 = 1; } while(0)
00714 #define SELECT_B_ResetPullup() do { WPUFbits.WPUF2 = 0; } while(0)
00715 #define SELECT_B_SetPushPull() do { ODCONFbits.ODCF2 = 0; } while(0)
00716 #define SELECT_B_SetOpenDrain() do { ODCONFbits.ODCF2 = 1; } while(0)
00717 #define SELECT_B_SetAnalogMode() do { ANSELFbits.ANSELF2 = 1; } while(0)
00718 #define SELECT_B_SetDigitalMode() do { ANSELFbits.ANSELF2 = 0; } while(0)
00719
00720 // get/set ENTER_B aliases
00721 #define ENTER_B_TRIS TRISFbits.TRISF3
00722 #define ENTER_B_LAT LATFbits.LATF3
00723 #define ENTER_B_PORT PORTFbits.RF3
00724 #define ENTER_B_WPU WPUFbits.WPUF3

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00725 #define ENTER_B_OD          ODCONFbits.ODCF3
00726 #define ENTER_B_ANS          ANSELFbits.ANSELF3
00727 #define ENTER_B_SetHigh()    do { LATFbits.LATF3 = 1; } while(0)
00728 #define ENTER_B_SetLow()     do { LATFbits.LATF3 = 0; } while(0)
00729 #define ENTER_B_Toggle()     do { LATFbits.LATF3 = ~LATFbits.LATF3; } while(0)
00730 #define ENTER_B_GetValue()   PORTFbits.RF3
00731 #define ENTER_B_SetDigitalInput() do { TRISFbits.TRISF3 = 1; } while(0)
00732 #define ENTER_B_SetDigitalOutput() do { TRISFbits.TRISF3 = 0; } while(0)
00733 #define ENTER_B_SetPullup()  do { WPUFbits.WPUF3 = 1; } while(0)
00734 #define ENTER_B_ResetPullup() do { WPUFbits.WPUF3 = 0; } while(0)
00735 #define ENTER_B_SetPushPull() do { ODCONFbits.ODCF3 = 0; } while(0)
00736 #define ENTER_B_SetOpenDrain() do { ODCONFbits.ODCF3 = 1; } while(0)
00737 #define ENTER_B_SetAnalogMode() do { ANSELFbits.ANSELF3 = 1; } while(0)
00738 #define ENTER_B_SetDigitalMode() do { ANSELFbits.ANSELF3 = 0; } while(0)
00739
00740 // get/set IO_RF4 aliases
00741 #define IO_RF4_TRIS          TRISFbits.TRISF4
00742 #define IO_RF4_LAT           LATFbits.LATF4
00743 #define IO_RF4_PORT          PORTFbits.RF4
00744 #define IO_RF4_WPU           WPUFbits.WPUF4
00745 #define IO_RF4_OD            ODCONFbits.ODCF4
00746 #define IO_RF4_ANS          ANSELFbits.ANSELF4
00747 #define IO_RF4_SetHigh()    do { LATFbits.LATF4 = 1; } while(0)
00748 #define IO_RF4_SetLow()     do { LATFbits.LATF4 = 0; } while(0)
00749 #define IO_RF4_Toggle()     do { LATFbits.LATF4 = ~LATFbits.LATF4; } while(0)
00750 #define IO_RF4_GetValue()   PORTFbits.RF4
00751 #define IO_RF4_SetDigitalInput() do { TRISFbits.TRISF4 = 1; } while(0)
00752 #define IO_RF4_SetDigitalOutput() do { TRISFbits.TRISF4 = 0; } while(0)
00753 #define IO_RF4_SetPullup()  do { WPUFbits.WPUF4 = 1; } while(0)
00754 #define IO_RF4_ResetPullup() do { WPUFbits.WPUF4 = 0; } while(0)
00755 #define IO_RF4_SetPushPull() do { ODCONFbits.ODCF4 = 0; } while(0)
00756 #define IO_RF4_SetOpenDrain() do { ODCONFbits.ODCF4 = 1; } while(0)
00757 #define IO_RF4_SetAnalogMode() do { ANSELFbits.ANSELF4 = 1; } while(0)
00758 #define IO_RF4_SetDigitalMode() do { ANSELFbits.ANSELF4 = 0; } while(0)
00759
00760 // get/set MAX_EN1 aliases
00761 #define MAX_EN1_TRIS          TRISFbits.TRISF5
00762 #define MAX_EN1_LAT           LATFbits.LATF5
00763 #define MAX_EN1_PORT          PORTFbits.RF5
00764 #define MAX_EN1_WPU           WPUFbits.WPUF5
00765 #define MAX_EN1_OD            ODCONFbits.ODCF5
00766 #define MAX_EN1_ANS          ANSELFbits.ANSELF5
00767 #define MAX_EN1_SetHigh()    do { LATFbits.LATF5 = 1; } while(0)
00768 #define MAX_EN1_SetLow()     do { LATFbits.LATF5 = 0; } while(0)
00769 #define MAX_EN1_Toggle()     do { LATFbits.LATF5 = ~LATFbits.LATF5; } while(0)
00770 #define MAX_EN1_GetValue()   PORTFbits.RF5
00771 #define MAX_EN1_SetDigitalInput() do { TRISFbits.TRISF5 = 1; } while(0)
00772 #define MAX_EN1_SetDigitalOutput() do { TRISFbits.TRISF5 = 0; } while(0)
00773 #define MAX_EN1_SetPullup()  do { WPUFbits.WPUF5 = 1; } while(0)
00774 #define MAX_EN1_ResetPullup() do { WPUFbits.WPUF5 = 0; } while(0)
00775 #define MAX_EN1_SetPushPull() do { ODCONFbits.ODCF5 = 0; } while(0)
00776 #define MAX_EN1_SetOpenDrain() do { ODCONFbits.ODCF5 = 1; } while(0)
00777 #define MAX_EN1_SetAnalogMode() do { ANSELFbits.ANSELF5 = 1; } while(0)
00778 #define MAX_EN1_SetDigitalMode() do { ANSELFbits.ANSELF5 = 0; } while(0)
00779
00780 // get/set IO_RF6 aliases
00781 #define IO_RF6_TRIS          TRISFbits.TRISF6
00782 #define IO_RF6_LAT           LATFbits.LATF6
00783 #define IO_RF6_PORT          PORTFbits.RF6
00784 #define IO_RF6_WPU           WPUFbits.WPUF6
00785 #define IO_RF6_OD            ODCONFbits.ODCF6
00786 #define IO_RF6_ANS          ANSELFbits.ANSELF6
00787 #define IO_RF6_SetHigh()    do { LATFbits.LATF6 = 1; } while(0)
00788 #define IO_RF6_SetLow()     do { LATFbits.LATF6 = 0; } while(0)
00789 #define IO_RF6_Toggle()     do { LATFbits.LATF6 = ~LATFbits.LATF6; } while(0)
00790 #define IO_RF6_GetValue()   PORTFbits.RF6
00791 #define IO_RF6_SetDigitalInput() do { TRISFbits.TRISF6 = 1; } while(0)
00792 #define IO_RF6_SetDigitalOutput() do { TRISFbits.TRISF6 = 0; } while(0)
00793 #define IO_RF6_SetPullup()  do { WPUFbits.WPUF6 = 1; } while(0)
00794 #define IO_RF6_ResetPullup() do { WPUFbits.WPUF6 = 0; } while(0)
00795 #define IO_RF6_SetPushPull() do { ODCONFbits.ODCF6 = 0; } while(0)
00796 #define IO_RF6_SetOpenDrain() do { ODCONFbits.ODCF6 = 1; } while(0)
00797 #define IO_RF6_SetAnalogMode() do { ANSELFbits.ANSELF6 = 1; } while(0)
00798 #define IO_RF6_SetDigitalMode() do { ANSELFbits.ANSELF6 = 0; } while(0)
00799
00800 // get/set IO_RF7 aliases
00801 #define IO_RF7_TRIS          TRISFbits.TRISF7
00802 #define IO_RF7_LAT           LATFbits.LATF7
00803 #define IO_RF7_PORT          PORTFbits.RF7
00804 #define IO_RF7_WPU           WPUFbits.WPUF7
00805 #define IO_RF7_OD            ODCONFbits.ODCF7
00806 #define IO_RF7_ANS          ANSELFbits.ANSELF7
00807 #define IO_RF7_SetHigh()    do { LATFbits.LATF7 = 1; } while(0)
00808 #define IO_RF7_SetLow()     do { LATFbits.LATF7 = 0; } while(0)
00809 #define IO_RF7_Toggle()     do { LATFbits.LATF7 = ~LATFbits.LATF7; } while(0)
00810 #define IO_RF7_GetValue()   PORTFbits.RF7
00811 #define IO_RF7_SetDigitalInput() do { TRISFbits.TRISF7 = 1; } while(0)

```



```

00812 #define IO_RF7_SetDigitalOutput()    do { TRISFbits.TRISF7 = 0; } while(0)
00813 #define IO_RF7_SetPullup()           do { WPUFbits.WPUF7 = 1; } while(0)
00814 #define IO_RF7_ResetPullup()         do { WPUFbits.WPUF7 = 0; } while(0)
00815 #define IO_RF7_SetPushPull()          do { ODCONFbits.ODCF7 = 0; } while(0)
00816 #define IO_RF7_SetOpenDrain()         do { ODCONFbits.ODCF7 = 1; } while(0)
00817 #define IO_RF7_SetAnalogMode()       do { ANSELFbits.ANSELF7 = 1; } while(0)
00818 #define IO_RF7_SetDigitalMode()       do { ANSELFbits.ANSELF7 = 0; } while(0)
00819
00830 void PIN_MANAGER_Initialize (void);
00831
00832
00833
00834
00835 #endif // PIN_MANAGER_H

```

4.22 pwm8.h

```

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00045 */
00046
00047 #ifndef PWM8_H
00048 #define PWM8_H
00049
00053
00054 #include <xc.h>
00055 #include <stdint.h>
00056
00057 #ifdef __cplusplus // Provide C++ Compatibility
00058     extern "C" {
00059
00060
00061     #endif
00062
00066
00067 #define PWM8_INITIALIZE_DUTY_VALUE    1
00068
00072
00102 void PWM8_Initialize(void);
00103
00129 void PWM8_LoadDutyValue(uint16_t dutyValue);
00130
00131
00132 #ifdef __cplusplus // Provide C++ Compatibility
00133     }
00134
00135     #endif
00136
00137
00138 #endif //PWM8_H

```

4.23 spi1.h

```

00001
00023

```

```

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00045 */
00046
00047 #ifndef SPI1_MASTER_H
00048 #define SPI1_MASTER_H
00049
00053
00054 #include <stdio.h>
00055 #include <stdint.h>
00056 #include <stdbool.h>
00057
00058 /* SPI interfaces */
00059 typedef enum {
00060     SPI1_DEFAULT
00061 } spil_modes_t;
00062
00063 void SPI1_Initialize(void);
00064 bool SPI1_Open(spil_modes_t spilUniqueConfiguration);
00065 void SPI1_Close(void);
00066 uint8_t SPI1_ExchangeByte(uint8_t data);
00067 void SPI1_ExchangeBlock(void *block, size_t blockSize);
00068 void SPI1_WriteBlock(void *block, size_t blockSize);
00069 void SPI1_ReadBlock(void *block, size_t blockSize);
00070 void SPI1_WriteByte(uint8_t byte);
00071 uint8_t SPI1_ReadByte(void);
00072
00073 #endif //SPI1_H

```

4.24 tmr2.h

```

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00045 */
00046
00047 #ifndef TMR2_H
00048 #define TMR2_H
00049
00053

```

```

00054 #include <stdint.h>
00055 #include <stdbool.h>
00056
00057 #ifdef __cplusplus // Provide C++ Compatibility
00058     extern "C" {
00059
00060 #endif
00061
00062
00066
00078
00079 typedef enum
00080 {
00081
00082     /* Roll-over Pulse mode clears the TMRx upon TMRx = PRx, then continue running.
00083     ON bit must be set and is not affected by Resets
00084     */
00085
00086     /* Roll-over Pulse mode indicates that Timer starts
00087     immediately upon ON = 1 (Software Control)
00088     */
00089     TMR2_ROP_STARTS_TMRON,
00090
00091     /* Roll-over Pulse mode indicates that the Timer starts
00092     when ON = 1 and TMRx_ers = 1. Stops when TMRx_ers = 0
00093     */
00094     TMR2_ROP_STARTS_TMRON_ERSHIGH,
00095
00096     /* Roll-over Pulse mode indicates that the Timer starts
00097     when ON = 1 and TMRx_ers = 0. Stops when TMRx_ers = 1
00098     */
00099     TMR2_ROP_STARTS_TMRON_ERSLOW,
00100
00101     /* Roll-over Pulse mode indicates that the Timer resets
00102     upon rising or falling edge of TMRx_ers
00103     */
00104     TMR2_ROP_RESETS_ERSBOTHEDGE,
00105
00106     /* Roll-over Pulse mode indicates that the Timer resets
00107     upon rising edge of TMRx_ers
00108     */
00109     TMR2_ROP_RESETS_ERSRISINGEDGE,
00110
00111     /* Roll-over Pulse mode indicates that the Timer resets
00112     upon falling edge of TMRx_ers
00113     */
00114     TMR2_ROP_RESETS_ERSFALLINGEDGE,
00115
00116     /* Roll-over Pulse mode indicates that the Timer resets
00117     upon TMRx_ers = 0
00118     */
00119     TMR2_ROP_RESETS_ERSLOW,
00120
00121     /* Roll-over Pulse mode indicates that the Timer resets
00122     upon TMRx_ers = 1
00123     */
00124     TMR2_ROP_RESETS_ERSHIGH,
00125
00126     /*In all One-Shot mode the timer resets and the ON bit is
00127     cleared when the timer value matches the PRx period
00128     value. The ON bit must be set by software to start
00129     another timer cycle.
00130     */
00131
00132     /* One shot mode indicates that the Timer starts
00133     immediately upon ON = 1 (Software Control)
00134     */
00135     TMR2_OS_STARTS_TMRON,
00136
00137     /* One shot mode indicates that the Timer starts
00138     when a rising edge is detected on the TMRx_ers
00139     */
00140     TMR2_OS_STARTS_ERSRISINGEDGE ,
00141
00142     /* One shot mode indicates that the Timer starts
00143     when a falling edge is detected on the TMRx_ers
00144     */
00145     TMR2_OS_STARTS_ERSFALLINGEDGE ,
00146
00147     /* One shot mode indicates that the Timer starts
00148     when either a rising or falling edge is detected on TMRx_ers
00149     */
00150     TMR2_OS_STARTS_ERSBOTHEDGE,
00151
00152     /* One shot mode indicates that the Timer starts
00153     upon first TMRx_ers rising edge and resets on all
00154     subsequent TMRx_ers rising edges

```

```

00155      */
00156      TMR2_OS_STARTS_ERSTIRSTRISINGEDGE,
00157
00158      /* One shot mode indicates that the Timer starts
00159      upon first TMRx_ers falling edge and restarts on all
00160      subsequent TMRx_ers falling edges
00161      */
00162      TMR2_OS_STARTS_ERSTIRSTFALLINGEDGE,
00163
00164      /* One shot mode indicates that the Timer starts
00165      when a rising edge is detected on the TMRx_ers,
00166      resets upon TMRx_ers = 0
00167      */
00168      TMR2_OS_STARTS_ERSTRISINGEDGEDETECT,
00169      /* One shot mode indicates that the Timer starts
00170      when a falling edge is detected on the TMRx_ers,
00171      resets upon TMRx_ers = 1
00172      */
00173      TMR2_OS_STARTS_ERSTFALLINGEDGEDETECT,
00174
00175      /* One shot mode indicates that the Timer starts
00176      when a TMRx_ers = 1, ON = 1 and resets upon TMRx_ers = 0
00177      */
00178      TMR2_OS_STARTS_TMRON_ERSHIGH = 0x16,
00179
00180      /* One shot mode indicates that the Timer starts
00181      when a TMRx_ers = 0, ON = 1 and resets upon TMRx_ers = 1
00182      */
00183      TMR2_OS_STARTS_TMRON_ERLOW = 0x17,
00184
00185      /*In all Mono-Stable mode the ON bit must be initially set, but
00186      not cleared upon the TMRx = PRx, and the timer will start upon
00187      an TMRx_ers start event following TMRx = PRx.*/
00188
00189      /* Mono Stable mode indicates that the Timer starts
00190      when a rising edge is detected on the TMRx_ers and ON = 1
00191      */
00192      TMR2_MS_STARTS_TMRON_ERSTRISINGEDGEDETECT = 0x11,
00193
00194      /* Mono Stable mode indicates that the Timer starts
00195      when a falling edge is detected on the TMRx_ers and ON = 1
00196      */
00197      TMR2_MS_STARTS_TMRON_ERSTFALLINGEDGEDETECT = 0x12,
00198
00199      /* Mono Stable mode indicates that the Timer starts
00200      when either a rising or falling edge is detected on TMRx_ers
00201      and ON = 1
00202      */
00203      TMR2_MS_STARTS_TMRON_ERSTBOTHEDGE = 0x13
00204
00205 } TMR2_HLT_MODE;
00206
00219
00220 typedef enum
00221 {
00222     /* T2INPPS is the Timer external reset source
00223     */
00224     TMR2_T2INPPS,
00225
00226     /* Reserved enum cannot be used
00227     */
00228     TMR2_RESERVED,
00229
00230     /* Timer4 Postscale is the Timer external reset source
00231     */
00232     TMR2_T4POSTSCALED,
00233
00234     /* Timer6 Postscale is the Timer external reset source
00235     */
00236     TMR2_T6POSTSCALED,
00237
00238     /* CCP1_OUT is the Timer external reset source
00239     */
00240     TMR2_CCP1_OUT,
00241
00242     /* CCP2_OUT is the Timer external reset source
00243     */
00244     TMR2_CCP2_OUT,
00245
00246     /* CCP3_OUT is the Timer external reset source
00247     */
00248     TMR2_CCP3_OUT,
00249
00250     /* CCP4_OUT is the Timer external reset source
00251     */
00252     TMR2_CCP4_OUT,
00253

```

```
00254     /* PWM5_out is the Timer external reset source
00255     */
00256     TMR2_PWM5_OUT,
00257
00258     /* PWM6_out is the Timer external reset source
00259     */
00260     TMR2_PWM6_OUT,
00261
00262     /* PWM7_out is the Timer external reset source
00263     */
00264     TMR2_PWM7_OUT,
00265
00266     /* PWM8_out is the Timer external reset source
00267     */
00268     TMR2_PWM8_OUT,
00269
00270     /* Reserved enum cannot be used
00271     */
00272     TMR2_RESERVED_2,
00273
00274     /* Reserved enum cannot be used
00275     */
00276     TMR2_RESERVED_3,
00277
00278     /* C1_OUT_SYNC is the Timer external reset source
00279     */
00280     TMR2_C1_OUT_SYNC,
00281
00282     /* C2_OUT_SYNC is the Timer external reset source
00283     */
00284     TMR2_C2_OUT_SYNC,
00285
00286     /* ZCD_Output is the Timer external reset source
00287     */
00288     TMR2_ZCD_OUTPUT,
00289
00290     /* CLC1_out is the Timer external reset source
00291     */
00292     TMR2_CLC1_OUT,
00293
00294     /* CLC2_out is the Timer external reset source
00295     */
00296     TMR2_CLC2_OUT,
00297
00298     /* CLC3_out is the Timer external reset source
00299     */
00300     TMR2_CLC3_OUT,
00301
00302     /* CLC4_out is the Timer external reset source
00303     */
00304     TMR2_CLC4_OUT,
00305
00306     /* UART1_rx_edge is the Timer external reset source
00307     */
00308     TMR2_UART1_RX_EDGE,
00309
00310     /* UART1_tx_edge is the Timer external reset source
00311     */
00312     TMR2_UART1_TX_EDGE,
00313
00314     /* UART2_rx_edge is the Timer external reset source
00315     */
00316     TMR2_UART2_RX_EDGE,
00317
00318     /* UART2_tx_edge is the Timer external reset source
00319     */
00320     TMR2_UART2_TX_EDGE
00321
00322 } TMR2_HLT_EXT_RESET_SOURCE;
00323
00324
00325
00329
00333 void TMR2_Initialize(void);
00366
00401 void TMR2_ModeSet(TMR2_HLT_MODE mode);
00402
00436 void TMR2_ExtResetSourceSet(TMR2_HLT_EXT_RESET_SOURCE reset);
00437
00465 void TMR2_Start(void);
00466
00494 void TMR2_StartTimer(void);
00495
00526 void TMR2_Stop(void);
00527
00558 void TMR2_StopTimer(void);
```

```

00559
00593 uint8_t TMR2_Counter8BitGet(void);
00594
00628 uint8_t TMR2_ReadTimer(void);
00629
00667 void TMR2_Counter8BitSet(uint8_t timerVal);
00668
00706 void TMR2_WriteTimer(uint8_t timerVal);
00707
00758 void TMR2_Period8BitSet(uint8_t periodVal);
00759
00810 void TMR2_LoadPeriodRegister(uint8_t periodVal);
00811
00848 bool TMR2_HasOverflowOccured(void);
00849
00850 #ifdef __cplusplus // Provide C++ Compatibility
00851     }
00852 }
00853 #endif
00854 #endif
00855
00856 #endif // TMR2_H

```

4.25 tmr3.h

```

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00045 */
00046
00047 #ifndef TMR3_H
00048 #define TMR3_H
00049
00053
00054 #include <stdbool.h>
00055 #include <stdint.h>
00056
00057 #ifdef __cplusplus // Provide C++ Compatibility
00058     extern "C" {
00059
00060
00061 #endif
00062
00063 #define TMR3_INTERRUPT_TICKER_FACTOR    200
00064
00068
00101 void TMR3_Initialize(void);
00102
00130 void TMR3_StartTimer(void);
00131
00162 void TMR3_StopTimer(void);
00163
00197 uint16_t TMR3_ReadTimer(void);
00198
00236 void TMR3_WriteTimer(uint16_t timerVal);
00237
00272 void TMR3_Reload(void);
00273
00311 void TMR3_StartSinglePulseAcquisition(void);
00312
00350 uint8_t TMR3_CheckGateValueStatus(void);

```

```

00351
00368 void TMR3_Callback(void);
00369
00386 void TMR3_SetInterruptHandler(void (* InterruptHandler)(void));
00387
00404 extern void (*TMR3_InterruptHandler)(void);
00405
00422 void TMR3_DefaultInterruptHandler(void);
00423
00424 #ifdef __cplusplus // Provide C++ Compatibility
00425     }
00426 }
00427
00428 #endif
00429
00430 #endif // TMR3_H

```

4.26 tmr5.h

```

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00045 */
00046
00047 #ifndef TMR5_H
00048 #define TMR5_H
00049
00053
00054 #include <stdbool.h>
00055 #include <stdint.h>
00056
00057 #ifdef __cplusplus // Provide C++ Compatibility
00058     extern "C" {
00059
00060
00061 #endif
00062
00063 #define TMR5_INTERRUPT_TICKER_FACTOR    200
00064
00068
00101 void TMR5_Initialize(void);
00102
00130 void TMR5_StartTimer(void);
00131
00162 void TMR5_StopTimer(void);
00163
00197 uint16_t TMR5_ReadTimer(void);
00198
00236 void TMR5_WriteTimer(uint16_t timerVal);
00237
00272 void TMR5_Reload(void);
00273
00311 void TMR5_StartSinglePulseAcquisition(void);
00312
00350 uint8_t TMR5_CheckGateValueStatus(void);
00351
00368 void TMR5_Callback(void);
00369
00386 void TMR5_SetInterruptHandler(void (* InterruptHandler)(void));
00387
00404 extern void (*TMR5_InterruptHandler)(void);

```

```

00405
00422 void TMR5_DefaultInterruptHandler(void);
00423
00424 #ifdef __cplusplus // Provide C++ Compatibility
00425     }
00426 }
00427
00428 #endif
00429
00430 #endif // TMR5_H

```

4.27 tmr6.h

```

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00046 */
00047 #ifndef TMR6_H
00048 #define TMR6_H
00049
00053
00054 #include <stdint.h>
00055 #include <stdbool.h>
00056
00057 #ifdef __cplusplus // Provide C++ Compatibility
00058     extern "C" {
00059
00060
00061 #endif
00062
00066
00078
00079 typedef enum
00080 {
00081
00082     /* Roll-over Pulse mode clears the TMRx upon TMRx = PRx, then continue running.
00083     ON bit must be set and is not affected by Resets
00084     */
00085
00086     /* Roll-over Pulse mode indicates that Timer starts
00087     immediately upon ON = 1 (Software Control)
00088     */
00089     TMR6_ROP_STARTS_TMRON,
00090
00091     /* Roll-over Pulse mode indicates that the Timer starts
00092     when ON = 1 and TMRx_ers = 1. Stops when TMRx_ers = 0
00093     */
00094     TMR6_ROP_STARTS_TMRON_ERSHIGH,
00095
00096     /* Roll-over Pulse mode indicates that the Timer starts
00097     when ON = 1 and TMRx_ers = 0. Stops when TMRx_ers = 1
00098     */
00099     TMR6_ROP_STARTS_TMRON_ERSLOW,
00100
00101     /* Roll-over Pulse mode indicates that the Timer resets
00102     upon rising or falling edge of TMRx_ers
00103     */
00104     TMR6_ROP_RESETS_ERSETHEDGE,
00105
00106     /* Roll-over Pulse mode indicates that the Timer resets

```



```

00107     upon rising edge of TMRx_ers
00108     */
00109     TMR6_ROP_RESETS_ERSRISINGEDGE,
00110
00111     /* Roll-over Pulse mode indicates that the Timer resets
00112     upon falling edge of TMRx_ers
00113     */
00114     TMR6_ROP_RESETS_ERSFALLINGEDGE,
00115
00116     /* Roll-over Pulse mode indicates that the Timer resets
00117     upon TMRx_ers = 0
00118     */
00119     TMR6_ROP_RESETS_ERLOW,
00120
00121     /* Roll-over Pulse mode indicates that the Timer resets
00122     upon TMRx_ers = 1
00123     */
00124     TMR6_ROP_RESETS_ERHIGH,
00125
00126     /*In all One-Shot mode the timer resets and the ON bit is
00127     cleared when the timer value matches the PRx period
00128     value. The ON bit must be set by software to start
00129     another timer cycle.
00130     */
00131
00132     /* One shot mode indicates that the Timer starts
00133     immediately upon ON = 1 (Software Control)
00134     */
00135     TMR6_OS_STARTS_TMRON,
00136
00137     /* One shot mode indicates that the Timer starts
00138     when a rising edge is detected on the TMRx_ers
00139     */
00140     TMR6_OS_STARTS_ERSRISINGEDGE ,
00141
00142     /* One shot mode indicates that the Timer starts
00143     when a falling edge is detected on the TMRx_ers
00144     */
00145     TMR6_OS_STARTS_ERSFALLINGEDGE ,
00146
00147     /* One shot mode indicates that the Timer starts
00148     when either a rising or falling edge is detected on TMRx_ers
00149     */
00150     TMR6_OS_STARTS_ERSBOTHEDGE,
00151
00152     /* One shot mode indicates that the Timer starts
00153     upon first TMRx_ers rising edge and resets on all
00154     subsequent TMRx_ers rising edges
00155     */
00156     TMR6_OS_STARTS_ERSFIRSTRISINGEDGE,
00157
00158     /* One shot mode indicates that the Timer starts
00159     upon first TMRx_ers falling edge and restarts on all
00160     subsequent TMRx_ers falling edges
00161     */
00162     TMR6_OS_STARTS_ERSFIRSTFALLINGEDGE,
00163
00164     /* One shot mode indicates that the Timer starts
00165     when a rising edge is detected on the TMRx_ers,
00166     resets upon TMRx_ers = 0
00167     */
00168     TMR6_OS_STARTS_ERSRISINGEDGEDETECT,
00169     /* One shot mode indicates that the Timer starts
00170     when a falling edge is detected on the TMRx_ers,
00171     resets upon TMRx_ers = 1
00172     */
00173     TMR6_OS_STARTS_ERSFALLINGEDGEDETECT,
00174
00175     /* One shot mode indicates that the Timer starts
00176     when a TMRx_ers = 1, ON =1 and resets upon TMRx_ers =0
00177     */
00178     TMR6_OS_STARTS_TMRON_ERHIGH = 0x16,
00179
00180     /* One shot mode indicates that the Timer starts
00181     when a TMRx_ers = 0, ON = 1 and resets upon TMRx_ers =1
00182     */
00183     TMR6_OS_STARTS_TMRON_ERLOW = 0x17,
00184
00185     /*In all Mono-Stable mode the ON bit must be initially set, but
00186     not cleared upon the TMRx = PRx, and the timer will start upon
00187     an TMRx_ers start event following TMRx = PRx.*/
00188
00189     /* Mono Stable mode indicates that the Timer starts
00190     when a rising edge is detected on the TMRx_ers and ON = 1
00191     */
00192     TMR6_MS_STARTS_TMRON_ERSRISINGEDGEDETECT = 0x11,
00193

```

```

00194      /* Mono Stable mode indicates that the Timer starts
00195         when a falling edge is detected on the TMRx_ers and ON = 1
00196      */
00197      TMR6_MS_STARTS_TMRON_ERSFALLINGEDGEDETECT = 0x12,
00198
00199      /* Mono Stable mode indicates that the Timer starts
00200         when either a rising or falling edge is detected on TMRx_ers
00201         and ON = 1
00202      */
00203      TMR6_MS_STARTS_TMRON_ERSBOTHEDGE = 0x13
00204 } TMR6_HLT_MODE;
00205
00206
00207
00208
00209
00210 typedef enum
00211 {
00212     /* T6INPPS is the Timer external reset source
00213     */
00214     TMR6_T6INPPS,
00215
00216     /* Timer2 Postscale is the Timer external reset source
00217     */
00218     TMR6_T2POSTSCALED,
00219
00220     /* Timer4 Postscale is the Timer external reset source
00221     */
00222     TMR6_T4POSTSCALED,
00223
00224     /* Reserved enum cannot be used
00225     */
00226     TMR6_RESERVED,
00227
00228     /* CCP1_OUT is the Timer external reset source
00229     */
00230     TMR6_CCP1_OUT,
00231
00232     /* CCP2_OUT is the Timer external reset source
00233     */
00234     TMR6_CCP2_OUT,
00235
00236     /* CCP3_OUT is the Timer external reset source
00237     */
00238     TMR6_CCP3_OUT,
00239
00240     /* CCP4_OUT is the Timer external reset source
00241     */
00242     TMR6_CCP4_OUT,
00243
00244     /* PWM5_out is the Timer external reset source
00245     */
00246     TMR6_PWM5_OUT,
00247
00248     /* PWM6_out is the Timer external reset source
00249     */
00250     TMR6_PWM6_OUT,
00251
00252     /* PWM7_out is the Timer external reset source
00253     */
00254     TMR6_PWM7_OUT,
00255
00256     /* PWM8_out is the Timer external reset source
00257     */
00258     TMR6_PWM8_OUT,
00259
00260     /* Reserved enum cannot be used
00261     */
00262     TMR6_RESERVED_2,
00263
00264     /* Reserved enum cannot be used
00265     */
00266     TMR6_RESERVED_3,
00267
00268     /* C1_OUT_SYNC is the Timer external reset source
00269     */
00270     TMR6_C1_OUT_SYNC,
00271
00272     /* C2_OUT_SYNC is the Timer external reset source
00273     */
00274     TMR6_C2_OUT_SYNC,
00275
00276     /* ZCD_Output is the Timer external reset source
00277     */
00278     TMR6_ZCD_OUTPUT,
00279
00280     /* CLC1_out is the Timer external reset source
00281     */
00282     TMR6_CLC1_OUT,
00283
00284
00285
00286
00287
00288
00289
00290
00291
00292

```

```

00293
00294     /* CLC2_out is the Timer external reset source
00295     */
00296     TMR6_CLC2_OUT,
00297
00298     /* CLC3_out is the Timer external reset source
00299     */
00300     TMR6_CLC3_OUT,
00301
00302     /* CLC4_out is the Timer external reset source
00303     */
00304     TMR6_CLC4_OUT,
00305
00306     /* UART1_rx_edge is the Timer external reset source
00307     */
00308     TMR6_UART1_RX_EDGE,
00309
00310     /* UART1_tx_edge is the Timer external reset source
00311     */
00312     TMR6_UART1_TX_EDGE,
00313
00314     /* UART2_rx_edge is the Timer external reset source
00315     */
00316     TMR6_UART2_RX_EDGE,
00317
00318     /* UART2_tx_edge is the Timer external reset source
00319     */
00320     TMR6_UART2_TX_EDGE
00321
00322
00323 } TMR6_HLT_EXT_RESET_SOURCE;
00324
00325
00329
00333
00365 void TMR6_Initialize(void);
00366
00401 void TMR6_ModeSet(TMR6_HLT_MODE mode);
00402
00436 void TMR6_ExtResetSourceSet(TMR6_HLT_EXT_RESET_SOURCE reset);
00437
00465 void TMR6_Start(void);
00466
00494 void TMR6_StartTimer(void);
00495
00526 void TMR6_Stop(void);
00527
00558 void TMR6_StopTimer(void);
00559
00593 uint8_t TMR6_Counter8BitGet(void);
00594
00628 uint8_t TMR6_ReadTimer(void);
00629
00667 void TMR6_Counter8BitSet(uint8_t timerVal);
00668
00706 void TMR6_WriteTimer(uint8_t timerVal);
00707
00758 void TMR6_Period8BitSet(uint8_t periodVal);
00759
00810 void TMR6_LoadPeriodRegister(uint8_t periodVal);
00811
00812
00829 void TMR6_CallBack(void);
00846 void TMR6_SetInterruptHandler(void (* InterruptHandler)(void));
00847
00864 extern void (*TMR6_InterruptHandler)(void);
00865
00882 void TMR6_DefaultInterruptHandler(void);
00883
00884
00885 #ifdef __cplusplus // Provide C++ Compatibility
00886
00887     }
00888
00889 #endif
00890
00891 #endif // TMR6_H

```

4.28 uart1.h

```

00001
00023
00024 /*

```

```
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00043     SOFTWARE.
00044
00045 */
00046
00047 #ifndef UART1_H
00048 #define UART1_H
00049
00050
00051 #include <xc.h>
00052 #include <stdbool.h>
00053 #include <stdint.h>
00054
00055 #ifdef __cplusplus // Provide C++ Compatibility
00056
00057     extern "C" {
00058
00059 #endif
00060
00061 #define UART1_DataReady  (UART1_is_rx_ready())
00062
00063 typedef union {
00064     struct {
00065         unsigned perr : 1;
00066         unsigned ferr : 1;
00067         unsigned oerr : 1;
00068         unsigned reserved : 5;
00069     };
00070     uint8_t status;
00071 }uart1_status_t;
00072
00073 extern volatile uint8_t uart1TxBufferRemaining;
00074 extern volatile uint8_t uart1RxCount;
00075
00076 void UART1_Initialize(void);
00077
00078 bool UART1_is_rx_ready(void);
00079
00080 bool UART1_is_tx_ready(void);
00081
00082 bool UART1_is_tx_done(void);
00083
00084 uart1_status_t UART1_get_last_status(void);
00085
00086 uint8_t UART1_Read(void);
00087
00088 void UART1_Write(uint8_t txData);
00089
00090 void UART1_Transmit_ISR(void);
00091
00092 void UART1_Receive_ISR(void);
00093
00094 void UART1_RxDataHandler(void);
00095
00096 void UART1_SetFramingErrorHandler(void (* interruptHandler)(void));
00097
00098 void UART1_SetOverrunErrorHandler(void (* interruptHandler)(void));
00099
00100 void UART1_SetErrorHandler(void (* interruptHandler)(void));
00101
00102 void (*UART1_RxInterruptHandler)(void);
00103
00104 void (*UART1_TxInterruptHandler)(void);
```

```

00536
00537
00538
00555 void UART1_SetRxInterruptHandler(void (* InterruptHandler)(void));
00556
00573 void UART1_SetTxInterruptHandler(void (* InterruptHandler)(void));
00574
00575
00576
00577 #ifdef __cplusplus // Provide C++ Compatibility
00578     }
00579 }
00580
00581 #endif
00582
00583 #endif // UART1_H

```

4.29 uart2.h

```

00001
00023
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00030
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00045 */
00046
00047 #ifndef UART2_H
00048 #define UART2_H
00049
00053
00054 #include <xc.h>
00055 #include <stdbool.h>
00056 #include <stdint.h>
00057
00058 #ifdef __cplusplus // Provide C++ Compatibility
00059     extern "C" {
00060
00061
00062 #endif
00063
00067
00068 #define UART2_DataReady (UART2_is_rx_ready())
00069
00073
00074 typedef union {
00075     struct {
00076         unsigned perr : 1;
00077         unsigned ferr : 1;
00078         unsigned oerr : 1;
00079         unsigned reserved : 5;
00080     };
00081     uint8_t status;
00082 }uart2_status_t;
00083
00087 extern volatile uint8_t uart2TxBufferRemaining;
00088 extern volatile uint8_t uart2RxCount;
00089
00093
00115 void UART2_Initialize(void);
00116
00163 bool UART2_is_rx_ready(void);
00164
00211 bool UART2_is_tx_ready(void);

```

```

00212
00258 bool UART2_is_tx_done(void);
00259
00306 uart2_status_t UART2_get_last_status(void);
00307
00355 uint8_t UART2_Read(void);
00356
00380 void UART2_Write(uint8_t txData);
00381
00401 void UART2_Transmit_ISR(void);
00402
00422 void UART2_Receive_ISR(void);
00423
00443 void UART2_RxDataHandler(void);
00444
00461 void UART2_SetFramingErrorHandler(void (* interruptHandler)(void));
00462
00479 void UART2_SetOverrunErrorHandler(void (* interruptHandler)(void));
00480
00497 void UART2_SetErrorHandler(void (* interruptHandler)(void));
00498
00499
00500
00517 void (*UART2_RxInterruptHandler)(void);
00518
00535 void (*UART2_TxInterruptHandler)(void);
00536
00537
00538
00555 void UART2_SetRxInterruptHandler(void (* InterruptHandler)(void));
00556
00573 void UART2_SetTxInterruptHandler(void (* InterruptHandler)(void));
00574
00575
00576
00577 #ifdef __cplusplus // Provide C++ Compatibility
00578     }
00579 }
00580
00581 #endif
00582
00583 #endif // UART2_H

```

4.30 mconfig.h

```

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00019  * TERMS.
00020  */
00021
00022 /*
00023  * File:
00024  * Author:
00025  * Comments:
00026  * Revision history:
00027  */
00028
00029 // This is a guard condition so that contents of this file are not included
00030 // more than once.
00031 #ifndef MCONFIG_H
00032 #define MCONFIG_H
00033
00034
00035 #include "mcc_generated_files/mcc.h"
00036 #include "vconfig.h"
00037
00038 extern const char *build_date, *build_time;

```

```

00039 extern V_data V;
00040
00041 void diversion_pwm_full(void);
00042 void diversion_pwm_set(uint16_t);
00043 bool set_back_light_off(bool);
00044
00045 #endif
00046

```

4.31 msg_text.h

```

00001 /*
00002  * File:    msg_text.h
00003  * Author:  root
00004  *
00005  * Created on July 15, 2019, 7:32 AM
00006  */
00007
00008 #ifndef MSG_TEXT_H
00009 #define MSG_TEXT_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014 #include "mconfig.h"
00015 #include "mydisplay.h"
00016
00017     typedef enum {
00018         display_message = 0,
00019         display_online,
00020         display_comm,
00021     } DISPLAY_TYPES;
00022
00023     const V_help T[] = {
00024     {
00025         .message = "commands 1",
00026         .display = "displays 1",
00027     },
00028     {
00029         .message = "commands 2",
00030         .display = "displays 2",
00031     },
00032     {
00033         .message = "commands 3",
00034         .display = "displays 3",
00035     },
00036     {
00037         .message = "commands 4",
00038         .display = "displays 4",
00039     },
00040     };
00041
00042 #ifdef __cplusplus
00043 }
00044 #endif
00045
00046 #endif /* MSG_TEXT_H */
00047

```

4.32 mydisplay.h

```

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```

```

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00020  */
00021
00022 /*
00023  * File:
00024  * Author:
00025  * Comments:
00026  * Revision history:
00027  */
00028
00029 // This is a guard condition so that contents of this file are not included
00030 // more than once.
00031 #ifndef XC_HEADER_TEMPLATE_H
00032 #define XC_HEADER_TEMPLATE_H
00033
00034 #include <xc.h> // include processor files - each processor file is guarded.
00035 #include <stdio.h>
00036 #include <string.h>
00037
00038 #include "vconfig.h"
00039 #include "eadog.h"
00040 #include "timers.h"
00041
00042 typedef struct D_data { // vterm control data structure
00043     char lcd[4][4][40];
00044     uint8_t vterm : 1;
00045     D_CODES last_info;
00046 } D_data;
00047
00048 void MyeaDogM_WriteStringAtPos(const uint8_t, const uint8_t, char *);
00049 uint8_t update_lcd(uint8_t);
00050 uint8_t set_vterm(const uint8_t);
00051 char * get_vterm_ptr(const uint8_t, const uint8_t);
00052 void vterm_dump(void);
00053 void vterm_sequence(void);
00054 inline D_CODES display_info(void);
00055 inline D_CODES display_help(void);
00056 bool help_button(void);
00057 bool check_help(const bool);
00058 D_CODES set_display_info(const D_CODES);
00059 D_CODES set_temp_display_help(const D_CODES);
00060
00061
00062 #ifdef __cplusplus
00063 extern "C" {
00064 #endif /* __cplusplus */
00065
00066     // TODO If C++ is being used, regular C code needs function names to have C
00067     // linkage so the functions can be used by the c code.
00068
00069 #ifdef __cplusplus
00070 }
00071 #endif /* __cplusplus */
00072
00073 #endif /* XC_HEADER_TEMPLATE_H */
00074

```

4.33 ringbufs.h

```

00001 /*
00002  * File:   ringbufs.h
00003  * Author: rootswitcDC circuit breakerh debounce icdebounce icfuse type T
00004  *
00005  * Created on July 25, 2015, 2:03 PM
00006  */
00007
00008 #ifndef RINGBUFS_H
00009 #define RINGBUFS_H
00010
00011 #ifdef __cplusplus
00012 extern "C" {
00013 #endif
00014
00015 #include "vconfig.h"
00016
00017 #define RBUF_SIZE    64
00018
00019     typedef struct ringBufS_t {
00020         uint8_t buf[RBUF_SIZE];
00021         uint8_t head;
00022         uint8_t tail;
00023         uint8_t count;
00024

```



```

00024     } ringBufS_t;
00025
00026     void ringBufS_init(volatile ringBufS_t *_this);
00027     int8_t ringBufS_empty(ringBufS_t *_this);
00028     int8_t ringBufS_full(ringBufS_t *_this);
00029     uint8_t ringBufS_get(ringBufS_t *_this);
00030     void ringBufS_put(ringBufS_t *_this, const uint8_t c);
00031     void ringBufS_put_dma(ringBufS_t *_this, const uint8_t c);
00032     void ringBufS_put_dma_cpy(ringBufS_t *, const char *, const uint8_t);
00033     void ringBufS_flush(ringBufS_t *_this, const int8_t clearBuffer);
00034
00035
00036 #ifdef __cplusplus
00037 }
00038 #endif
00039
00040 #endif /* RINGBUFS_H */
00041

```

4.34 tests.h

```

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00019  * TERMS.
00020  */
00021
00022 /*
00023  * File:
00024  * Author:
00025  * Comments:
00026  * Revision history:
00027  */
00028
00029 // This is a guard condition so that contents of this file are not included
00030 // more than once.
00031 #ifndef TESTS_H
00032 #define TESTS_H
00033
00034 #include <xc.h> // include processor files - each processor file is guarded.
00035 #include "mcc_generated_files/pin_manager.h"
00036
00037 #define DEBUG_MPMC // global application debug switch
00038 #define DLED1     DEBUG1_LAT
00039 #define DLED2     DEBUG2_LAT
00040
00041 #ifndef DEBUG_MPMC
00042 #define DEBUG_JSON
00043 // #define DEBUG_DAQ1 // daq debug switches : SCANNER
00044 // #define DEBUG_DAQ2 // ADC conversion
00045 // #define DEBUG_SWH1 // switches
00046 // #define DEBUG_SWH2
00047 // #define DEBUG_BSOC1 // battery run calculations
00048 // #define DEBUG_BSOC2
00049 // #define DEBUG_HIST1
00050 // #define DEBUG_TMR3 // timers
00051 // #define DEBUG_TRM5
00052 // #define DEBUG_DISP1 // LCD Display
00053 // #define DEBUG_DISP2
00054 // #define DEBUG_WDT1 // WDT
00055 // #define DEBUG_WDT2
00056 #endif
00057
00058 #define CALIB     false
00059 #define FIXUPS     true
00060 // #define CLR_CYCLE_DATA
00061 // #define CLR_BATTERY_DATA
00062 // #define CLR_PV_DATA

```

```
00063 #endif /* XC_HEADER_TEMPLATE_H */
00064
```

4.35 timers.h

```
00001 #ifndef TIMERS_H
00002 #define TIMERS_H
00003
00004 #include <stdint.h>
00005 #include <stdbool.h>
00006
00007 //Software timers - use these to refer to timers rather than
00008 //integers. Add more as needed.
00009 //NOTE: 32767 ms max delay
00010
00011 enum APP_TIMERS {
00012     TMR_INTERNAL = 0, //Used in timers.c - do not remove or use elsewhere
00013     TMR_ADC,
00014     TMR_T2,
00015     TMR_T3,
00016     TMR_T4,
00017     TMR_MC_TX,
00018     TMR_HBIO,
00019     TMR_INFO,
00020     TMR_HELP,
00021     TMR_HELPDIS,
00022     TMR_DISPLAY,
00023     TMR_FLIPPER,
00024     TMR_ESR,
00025     TMR_TEXT,
00026     TMR_SPIN,
00027     TMR_GTI,
00028     //
00029     //(Add timers here as needed)
00030     //
00031     TMR_COUNT //number of timers - always keep at end of enum!
00032 };
00033
00034 void StartTimer(const uint8_t timer, const uint16_t count);
00035 bool TimerDone(const uint8_t timer);
00036 void WaitMs(const uint16_t numMilliseconds);
00037 void sw_timers_isr(void);
00038
00039 #endif //TIMERS_H
```

4.36 vconfig.h

```
00001 /* * File: vconfig.h
00002 * Author: root
00003 *
00004 * Created on February 19, 2019, 4:41 PM
00005 */
00006
00007 #ifndef VCONFIG_H
00008 #define VCONFIG_H
00009
00010 #include "mcc_generated_files/adcc.h"
00011 #include "mcc_generated_files/ext_int.h"
00012
00013 #ifdef __cplusplus
00014 extern "C" {
00015 #endif
00016
00017 #include <xc.h>
00018 #include <time.h>
00019 #include "mcc_generated_files/spi1.h"
00020 #include "mcc_generated_files/pin_manager.h"
00021 #include "ringbufs.h"
00022
00023 #define VER "2.20"
00024 /*
00025 * REMOVE THE CHIP PROGRAMMER, IT CHANGES THE ADC READINGS
00026 */
00027
00028 /*
00029 * 0.1 MBMC new version for one 24vdc battery string for the 57K42
00030 * 0.2 start to configure the hardware for mbmc duty
00031 * 0.3 configure for mbmc_io board
```

```

00032      * 0.4 ADC setup
00033      * 0.5 remove gemsecs coding
00034      * 0.6 adc scanning by threshold interrupt using burst average mode
00035      * 0.7 system data conversion from ADC data
00036      * 0.8 basic current and voltage tracking
00037      * 0.9 auto convert bitmaps
00038      * 0.91 start adding 32 utc time code
00039      * 0.94 switch input code
00040      * 0.95 MAX6818 interrupt coding
00041      * 0.97 need hardware fix on mbmc_io board to repair bypassed resistors R20 and R18
00042      * 0.98 last of the hardware bugs fixed
00043      * 0.99 power load calculations
00044      * 1.0+ mainly working code base
00045      * 1.07 move MAX6818 enable pin for board rev 1.1
00046      * 1.08 NHD 4x20 display
00047      * 1.16 update logging format
00048      * 1.17 current sensor calibration functions
00049      * 1.18 EEPROM routines
00050      * 1.19 8-bit crc for EEPROM change detection
00051      * 1.20 optional 10A calibration
00052      * 1.21 history tracking in EEPROM
00053      * 1.21 history update in EEPROM
00054      * 1.23 line 4 display issues with history display functions, temp fix is to set MAX_LINE to 19
00055      *      need to work on this for a correct fix. review LCD_addressing.pdf in host_board folder
00056      * 1.24 fix LCD problems, was trying to send a 0x00 using printf
00057      * 1.25 load time_t date from saved EEPROM history if valid
00058      * 1.29 fine tuning the ADC configuration
00059      * 1.31 add third hall-sensor for actual PV current and CC current for MPPT VI conversion tracking
00060      * 1.32 data number fixups for display and calculations
00061      * 1.33 reorder logging data format
00062      * 1.34 CC mode to voltage decoding routines
00063      * 1.35 AC charger limits and float cycle counts for H[11]
00064      * 1.36 Charge controller op mode FIXUPS
00065      * 1.37 HID button response fixes, use bitmap data for stored button press info in hid.c
00066      * 1.38 UART 1 serial control parser. 'V' AC charger ON, 'v' AC charger off, '#' Show system
information
00067      * 1.39 Tweak one second interrupt times for better precision
00068      * 1.40 SOC and AH corrections at FLOAT and BOOST triggers from charge controller
00069      * 1.41 add daily ah tracking
00070      * 1.42 fix daily tracking
00071      * 1.43 add option using BAT_100A in daq.h to select Honeywell sensor for battery current sensor
00072      * 1.44 DC dump load from PV functions
00073      * 1.45 WDT hang protection
00074      * 1.46 server time updates
00075      * 1.47 Battery low voltage checks for charger activation
00076      * 1.48 add third current sensor zero calibration
00077      * 1.49 restructure and optimizations
00078      * 1.50 current sensor options in daq.h: CAL_10A CAL_TWO
00079      * 1.51 change ADC to FRC clock source
00080      * 1.52 back to ext clock for ADC with increased time for hold sample
00081      * 1.53 fix lcd display line limits
00082      * 1.56 connection updates and structure fixes
00083      * 2.00 FM80 dump load system monitor
00084      * 2.01 swap around the voltage and current sensors offsets and scalars for the FM80 system
configuration
00085      * 2.02 move variables on the LCD display code to match the actual measurements
00086      * 2.03 add json power variables, change MQTT data update to 1 second
00087      * 2.04 general code cleanup
00088      * 2.05 GTI power control and monitoring using UART2
00089      * 2.08 GTI command set, serial UART2
00090      * 2.09 Fix UART1 receive bugs
00091      * 2.10 Alpha version
00092      * 2.11 typo corrections
00093      * 2.12 cleanup GTI parser code
00094      * 2.13 add GTI power value parsing with V start value, [0..9][3], X execute value
00095      * 2.14 add cleanup
00096      * 2.15 more cleanups
00097      * 2.16 add charge controller mode to MQTT variables, fix CCMODE voltage ranges
00098      * 2.17 add UTC time setting from MQTT
00099      * 2.18 doc updates and minor fixes
00100      * 2.19 current and power fixups for FM80 coding to JSON
00101      * 2.20 update to use xc8 3.0
00102      */
00103
00104      /*
00105      * REMOVE THE CHIP PROGRAMMER, IT CHANGES THE ADC READINGS
00106      */
00107      //#define TESTING
00108      //#define DISPLAY_SLOW
00109      #define DEF_TIME      1721693000
00110      #define DEF_TIME_SIZE  11
00111
00112      #define SLED      LED0_LAT
00113
00114      #define EADOGM_CMD_CLR      1
00115      #define EADOGM_CMD_CURSOR_ON      0b00001111
00116      #define EADOGM_CMD_CURSOR_OFF      0b00001100

```

```

00117 #define EADOGM_CMD_DISPLAY_ON      0b00001100
00118 #define EADOGM_CMD_DISPLAY_OFF      0b00001000
00119 #define EADOGM_CMD_DDRAM_ADDR       0b10000000
00120 #define EADOGM_CMD_CGRAM_ADDR       0b01000000
00121 #define EADOGM_CMD_SELECT_R0        0b00011000
00122 #define EADOGM_CMD_SELECT_R1        0b00010000
00123 #define EADOGM_CMD_SET_TABLE2       0b00101010
00124 #define EADOGM_COLSPAN              16
00125
00126 #define NHD          // 4x20 SPI display
00127
00128 #define T1  500
00129 #define T2  3000
00130 #define T3  5000
00131 #define T4  5000
00132 #define HBTS 5000 // short ping delay
00133 #define HBTL 30000 // ping delay
00134 #define RTY 3
00135 #define ERROR_CHECKSUM 31500
00136 #define ERROR_COMM 31500
00137
00138 #define DEFAULT_TID 1
00139 #define TDELAY 3000
00140 #define LDELAY 1000
00141 #define SDELAY 500
00142 #define BDELAY 300
00143 #define DDELAY 125 // display update spacing 125
00144 #define TXTDELAY 10000 // unix time from server timeout
00145 #define SOCDELAY 250 // sync SPI and UART dma 250
00146 #define DFLIP 1500 // display info flipping spacing
00147
00148 #define LOG_WAIT 1 // data logging interval in seconds
00149 #define D_CODE '5' // data logging line code version
00150 #define I_CODE '*' // info logging line code
00151
00152 #define CAL_DELAY 64
00153 // #define SKIP_ESR
00154
00155 /*
00156  * offsets in bytes
00157  */
00158
00159 #define TX_RESERVE 59
00160 #ifndef NHD
00161 #define MAX_LINE 20
00162 #else
00163 #define MAX_LINE 16
00164 #endif
00165 #define MAX_BUTTONS 8
00166
00167 #define HPARAM_SIZE 13
00168 #define HVER 1
00169
00170 #define BL_TIME 600
00171
00172 #define CC_DEACT 74 // 1.10
00173 #define CC_ACT 100 // 1.43
00174 #define CC_MPPT 120 // 1.75
00175 #define CC_EQUAL 140 // 2.10
00176 #define CC_BOOST 160 // 2.45
00177 #define CC_FLOAT 180 // 2.82
00178 #define CC_LIMIT 200 // 3.20
00179 #define CC_OFFLINE 240 // 4.00
00180
00181 #define max_port_data 1024
00182
00183 struct spi_link_type { // internal SPI state table
00184     uint8_t SPI_LCD : 1;
00185     uint8_t SPI_AUX : 1;
00186     uint8_t LCD_TIMER : 1;
00187     volatile uint8_t LCD_DATA : 1;
00188     uint16_t delay;
00189     uint8_t config;
00190     uint8_t * txbuf;
00191     volatile int32_t int_count;
00192 };
00193
00194 typedef enum {
00195     DIS_STR = 0,
00196     DIS_HELP,
00197     DIS_ERR,
00198     DIS_CLEAR,
00199 } D_CODES;
00200
00201 typedef struct terminal_type {
00202     uint8_t msgid;
00203     uint8_t TID, mcode, mparm, cmdlen, log_seq;

```

```

00204     uint8_t host_display_ack : 1;
00205     D_CODES info, help_temp;
00206     uint16_t ceid;
00207     uint16_t log_num;
00208 } terminal_type;
00209
00210 typedef enum {
00211     UI_STATE_INIT = 0,
00212     UI_STATE_HOST,
00213     UI_STATE_DEBUG,
00214     UI_STATE_LOG,
00215     UI_STATE_MON,
00216     UI_STATE_ERROR
00217 } UI_STATES;
00218
00219 typedef enum {
00220     SW_OFF = 0,
00221     SW_ON,
00222     SW_INVALID,
00223 } SW_STATES;
00224
00225 typedef enum {
00226     S0 = 0,
00227     S1 = 1,
00228     SSELECT = 2,
00229     SENTER = 3,
00230     SCALIB = 4,
00231     S5,
00232     S6,
00233     SNULL,
00234 } SW_NAMES;
00235
00236 typedef struct V_data { // control data structure
00237     UI_STATES ui_state;
00238     char buf[64], info[64], rbuf[16];
00239     volatile time_t ticks, blight, ac_time;
00240     volatile int8_t testing;
00241     volatile uint8_t sys_info : 1, ac_on : 1, ac_off : 1, time_info : 1, get_time_text : 1;
00242     uint8_t error, abort, msg_error, msg_ret, alarm, float_ticks, boost_ticks, wdt_ticks;
00243     UI_STATES ui_sw;
00244     uint16_t r_checksum, t_checksum, checksum_error, mode_pwm, sequences, all_errors;
00245     uint8_t set_sequ : 1, debug : 1, help : 1, stack : 3, help_id : 2, screen : 1;
00246     terminal_type response;
00247     volatile uint8_t ticker, cc_state;
00248     bool flipper, calib, enter, sensor_set, fixup, cc_active;
00249     volatile bool system_stable, blight_off, in_float, in_boost;
00250     volatile uint32_t lowint_count, timerint_count, tx_count, rx_count, adc_count, spi_count,
switch_count;
00251 } V_data;
00252
00253 typedef struct V_help {
00254     const char message[22], display[22];
00255 } V_help;
00256
00257 /*      hist[x].h[x]
00258 *      h0 Real Ah usage this cycle
00259 *      h1 Total charge cycles into at least boost
00260 *      h2 Current lowest discharge,
00261 *      h3 PV ah total
00262 *      h4 Batt W out
00263 *      h5 Batt W in
00264 *      h6 Real Ah usage
00265 *      h7 Batt Voltage full load
00266 *      h8 Batt Voltage one load
00267 *      h9 Lowest ESR
00268 *      h10 Highest ESR
00269 *      h11 Total charge cycles into float
00270 *      h12 Total kWh in
00271 */
00272
00273 typedef struct hist_type {
00274     int32_t pv_eff, tot_eff; // pv generation eff factor, total system eff factor
00275     uint32_t ttg_t, updates;
00276     time_t pclock;
00277     float peukert, cef, peukert_adj, cef_calc, cef_save;
00278     int16_t h[HPARAM_SIZE]; // h[6]=cumulative battery Ah cc and inv (real),h[0]=cumulative
battery Ah cc and inv (p_adj)
00279     uint16_t rate, udod, bsoc, bound_rate, bound_factor, samplei, sampleo, ah, drate, esr,
rest_rate, rest_factor, esrp;
00280     uint8_t version;
00281 } hist_type;
00282
00283 #ifdef __cplusplus
00284 }
00285 #endif
00286
00287 #endif /* VCONFIG_H */

```

00288

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